

FORECASTING CUSTOMERS' RISK-ADJUSTED REVENUE: AN EXPLAINABLE MACHINE LEARNING APPROACH FOR THE TELECOMMUNICATION INDUSTRY

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Businesses increasingly rely on Customer Lifetime Value (CLV) metrics to inform strategic customer engagement and retention strategies. However, a systematic literature review reveals a significant gap in the integration of customers' risk factors into CLV calculations. Despite the large amount of customer data collected by companies, risk-adjusted CLV predictions using Machine Learning (ML) have been largely underexplored. This study addresses this gap by proposing a novel set of Risk-Adjusted Return (RAR) metrics tailored to noncontractual (B2C) settings in the telecommunications industry. Using the Cross-Industry Standard Process for Data Mining (CRISP-DM), ML models are designed to incorporate customer churn probability and beta value into the discount rate for CLV calculations. Four RAR metrics are introduced and validated using eXplainable Artificial Intelligence (XAI) techniques. The results demonstrate the distinctiveness of the RAR approaches, with high accuracy in churn prediction (85%) and strong RAR model performance (R^2 of 92% and MAPE of 20%). XGBoost shows superior performance in churn prediction, while CatBoost excels in RAR prediction. Key features influencing RAR include loyalty points, revenue metrics, churn probability, and beta value, consistent with traditional RAR calculation factors.

Keywords: Risk-adjusted revenue; machine learning; eXplainable AI; telecommunication industry.

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1. Introduction

Customers are the backbone of any business, constituting not only the primary revenue source but also a core asset influencing companies' performance, profitability, and long-term growth trajectory. This has precipitated a shift in corporate paradigms toward a more customer-centric approach, heralding the advent of Relationship Marketing (RM) in the 1980s. RM endeavors to attract, retain, and fortify customer relationships, emphasizing the enduring value of customer connections [35].

Central to the ethos of RM is the imperative for companies to ascertain the worth of their clientele and craft strategies to nurture and retain them. A widely embraced metric for gauging customer value is Customer Lifetime Value (CLV) [12, 13, 19]. CLV encapsulates the total net present value of all anticipated future cash flows generated by a customer over their engagement with a company, net of the costs associated with acquiring and servicing that customer [13, 19].

By envisaging a customer's prospective value, companies can make judicious decisions regarding customer management, delineating between customers of significant value warranting retention and those deemed less valuable. Furthermore, the strategic deployment of CLV enables companies to enhance profitability by effectively targeting high-value customers, thus fostering enduring and mutually beneficial relationships [7, 19, 31]. The identification and prioritization of high-value customers empower companies to allocate resources judiciously, leveraging tailored marketing initiatives [2], loyalty programs [17], and personalized offerings [8] to bolster retention and augment CLV. Consequently, bolstering customer retention and value can catalyze heightened profitability and sustainable organizational success.

Moreover, CLV serves not only as a barometer of customer worth but also as a reflection of a company's financial performance and valuation [13, 14]. Integrating CLV with projected customer growth enables companies to estimate their present valuation accurately. Given that customer value constitutes a substantial fraction of a firm's overall valuation, CLV emerges as a salient proxy for firm worth [13]. Notably, companies with elevated CLV typically command higher stock prices, underscoring investors' confidence in their future earnings potential [33].

Nevertheless, despite its paramount importance, the conventional computation of CLV fails to capture the dynamic nature of markets, presupposing that companies remain static post-customer acquisition. This oversight disregards the potential ramifications of evolving customer preferences, such as churn or diminished purchases, which pose risks to future profitability. Consequently, attaining an accurate estimation of customer value mandates the integration of associated risks into the calculation [6, 49].

While numerous studies acknowledge the significance of integrating customer risk into CLV calculations, research focusing specifically on the Business-to-Consumer (B2C) sector, particularly within the noncontractual customer segment of the telecommunications industry, remains scarce. This study endeavors to fill this gap by

developing a bespoke risk-adjusted CLV metric tailored to the B2C landscape, specifically within the telecommunications sector. Moreover, while some studies harness Machine Learning (ML) algorithms to predict risk-adjusted CLV, their application in the telecommunications industry remains nascent. This research aspires to introduce a pioneering metric that encapsulates risk's impact on customer value, concentrating on telecommunication industry applications, and explore the potential of ML algorithms in predicting risk-adjusted CLV for heightened accuracy and precision. Through these endeavors, this study seeks to provide deeper insights into customer value dynamics and equip telecommunication companies with the tools to make informed decisions regarding customer acquisition and retention strategies.

The rest of the paper is structured as outlined below: Sec. 2 includes the review of the relevant literature. Sections 3 and 4 outline the suggested methodology and outcomes of the empirical study. Section 5 discusses the final concerns, emphasizing the study's limitations and suggesting areas for future research.

2. Related Work

2.1. *Customer value and methods to incorporate customers' risk in CLV models*

The synthesis of existing literature reveals a prevalent reliance on financial portfolio theory in studies incorporating customer risk into CLV computations. This body of research draws extensively from the foundational contributions of Nobel Laureate William Sharpe, notably his association with Modern Portfolio Theory (MPT) and the Capital Asset Pricing Model (CAPM) [37], as well as Harry Markowitz's seminal work on portfolio selection theory [27]. MPT advocates maximizing investor returns for a quantified level of risk through diversified share portfolios, while CAPM, predicated on risk aversion, posits that investors demand higher returns for greater risk levels. CAPM distinguishes between systematic, affecting all market assets, and unsystematic risk, specific to individual assets. It asserts that while unsystematic risk can be mitigated through diversification, systematic risk, quantified by beta, remains inherent [6, 37].

In the realm of customer valuation, studies modeling CLV with risk considerations often treat customers as assets with varying inherent risk levels, necessitating distinct management strategies to optimize returns. Noteworthy among these are the works of Dhar and Glazer [9] and Ryals [34], which apply CAPM for computing customer-specific discount rates in CLV models. However, critiques by scholars like Wangenheim and Lentz [47] and Buhl and Heinrich [6] highlight CAPM's shortcomings, particularly its omission of unsystematic risk. To address these limitations, these researchers advocate for the application of Markowitz's portfolio selection theory to customer portfolio management, citing its efficacy in mitigating such constraints.

The adaptation of customer portfolio theory to dynamic customer segment theory has also gained traction, as evidenced by Homburg *et al.* [15]. They argue that static models may misjudge the value of different customer tiers by neglecting the dynamic nature of customer relationships. This perspective is further elaborated by Viviani *et al.* [46] in the context of the hospitality industry.

Other scholars have extended the application of portfolio selection theory to various customer portfolio management scenarios across sectors such as e-commerce, B2B, and financial services, with studies like those by Sackmann *et al.* [36], Tarasi *et al.* [42] and Juhl and Christensen [16] being exemplary. Norouzi and Albadv [28] innovated in this area by integrating stochastic CLV modeling with *ex-ante* customer portfolio optimization, predicting future customer returns, and incorporating these into portfolio optimization.

Machado and Karray [25] propose a novel adaptation of the portfolio selection theory model, utilizing ML to categorize customers and predict Risk-Adjusted Return (RAR) values. These diverse studies collectively underscore the applicability and value of financial portfolio theory in managing customer portfolios.

An alternative approach, as outlined in studies like those by Machado and Karray [25] and Singh *et al.* [38], considers Multiple Sources of Risk (MSR) relevant to specific industries and business contexts. These studies acknowledge the limited scope of traditional risk models, which typically focus on singular risk types like income volatility. Consequently, they propose frameworks encompassing multiple risk types, including customer churn probability and volatility of different income sources.

The industrial settings wherein customer risk incorporation into CLV is explored can be broadly categorized into three sectors: financial services, B2C outside financial services, and B2B. The majority of research is found within financial services, encompassing insurance, credit card industries, and peer-to-peer lending. B2C settings, including e-commerce, telecommunications, and airlines, account for a substantial proportion of the studies, while B2B settings, such as medical instruments for hospitals and commodities companies, comprise the remaining fraction.

These studies predominantly focus on optimizing customer portfolio composition to maximize returns, as evidenced by the prevalence of “customer portfolio management” as a key research area. The applications span various industries, from financial services to B2C and B2B sectors, with specific foci such as identifying profitable customers in credit card industries, adjusting product pricing in e-commerce, and developing more accurate risk-adjusted CLV metrics.

The integration of customer risk into valuation depends on the research’s specific application area. Studies applying financial portfolio concepts typically use mean–variance analysis for optimal customer portfolio composition. These studies often segment customers based on demographic characteristics before employing the model for portfolio optimization.

For example, Tarasi *et al.* [42] employed mean–variance analysis after segmenting customers based on market segments, while Viviani *et al.* [46] used the Hidden

Markov Model (HMM) for customer segmentation, considering segment switching probabilities as a risk factor. Apart from mean–variance analysis, other methods like particle swarm optimization are also employed for optimal portfolio construction.

In contrast, other studies with varied application areas adopt different methodologies. Ryals and Knox [35] employed a mathematical approach for the direct calculation of risk-adjusted CLV, while So *et al.* [41] utilized statistical models for predictive analysis. Similarly, Albadvi and Norouzi [1] combined the RFM model with mathematical approaches for risk calculation, and Singh *et al.* [38, 39] used Data Envelopment Analysis (DEA) for RAR computation in credit card companies.

2.2. Machine learning in customer value assessments

The analysis of existing literature underscores a limited application of ML algorithms in predicting risk-adjusted CLV, with only four notable studies in this domain. Wangenheim and Lentz [47] conducted a pioneering study employing multiple linear regression to forecast customer revenue post risk-adjustment, followed by customer segmentation via *k*-Means clustering. Similarly, Homburg *et al.* [15] integrated ML algorithms, particularly a regression tree, for customer data segmentation, subsequently utilizing Mean–Variance optimization for customer portfolio management.

A more recent study by Machado and Karray [25] explored the application of ML techniques for predicting customers' RAR within the financial lending sector. This study innovatively incorporated the Probability of Default (PD) of lenders, derived through logistic regression, as a significant risk factor. The researchers employed credit ratings as a predictive tool for lender PD, employing distinct encoding methods and comparing outcomes. These methodologies signify a notable advancement in integrating diverse risk sources into CLV calculations, highlighting the synergistic potential of traditional statistical methods with ML techniques.

Building upon this foundation, Machado and Karray [24] conducted further research representing the comprehensive utilization of ML for predicting risk-adjusted CLV. They introduced a hybrid ML framework for estimating RAR values of customer segments, combining supervised and unsupervised learning algorithms to leverage their respective strengths and address individual model limitations. Hybrid models excel in managing high-dimensional datasets and amalgamating beneficial attributes of constituent models.

In their study, Machado and Karray [24] compared outcomes of individual and hybrid models using data from the peer-to-peer lending industry. The hybrid approach involved initial customer data clustering using unsupervised learning algorithms, followed by employing clustering results as variables in supervised learning models to predict customer RAR values. The study evaluated various supervised learning algorithms, including Adaptive Boosting (AB), Gradient Boosting (GB), Decision Tree (DT), Random Forest (RF), Support Vector Machine (SVM), and Artificial Neural Network (ANN). Empirical results indicated that hybrid ML models generally outperformed individual models in predictive accuracy and

efficiency. Combinations of DT, RF, or GB with k -Means++ exhibited superior performance compared to standalone counterparts. Consequently, research by Machado and Karray [24] illuminates the substantial potential of hybrid ML models in enhancing the accuracy of risk-adjusted CLV predictions, offering valuable insights for future scholarly exploration in this field.

2.3. The gap in the literature: Highlighting the contributions of this study

The current landscape of research in customer valuation reveals several notable gaps warranting further exploration and analysis. Firmansyah *et al.* [10] presented a systematic review of the applications of AI in assessing customer value and risk-adjusted revenue. Their findings support our claims regarding the gaps identified in the literature. While they acknowledge the existence of traditional methods for assessing CLV and RAR, the authors also emphasize the largely unexplored area of empirical studies using ML to assess CLV and RAR. A primary deficiency lies in the disproportionate concentration of studies within the Financial Services Industry (FSI), with a glaring underrepresentation of other sectors, notably the B2C domain. This imbalance is particularly striking given the unique characteristics and challenges inherent to non-FSI industries. For instance, the telecommunications sector, known for its high customer acquisition costs and low switching barriers, remains significantly underexplored in terms of integrating customer risk into valuation models. This oversight is significant, considering the potential insights that customer valuation in telecommunications could offer for optimizing acquisition and retention strategies.

Another critical gap identified relates to the underutilization of ML algorithms in customer valuation. Despite the increasing application of ML in this field, there remains a stark disparity in studies that fully harness ML for predicting the impact of customer risk. While numerous studies focus on assessing the influence of customer risk on CLV, only a few comprehensively employ ML algorithms for predicting risk-adjusted customer value. Given the vast availability of big data and advanced computational resources, ML holds immense potential for delivering robust predictions about customer behavior and associated risks. Therefore, a comparative analysis between ML and traditional methods in calculating risk-adjusted customer value could yield invaluable insights into their respective efficacies and applicability across diverse industry landscapes, aiding organizations in making more informed decisions regarding customer strategies.

The third notable research gap concerns the application of eXplainable Artificial Intelligence (XAI) in customer valuation. Despite growing interest in XAI, only a solitary study in this field has been identified, with no research found applying XAI in the context of risk-adjusted revenue prediction. This gap is critical considering the growing importance of ML model explainability due to interpretability, regulatory, and ethical implications. Integrating XAI into risk-adjusted revenue prediction could

significantly enhance the interpretability of ML models, leading to more accurate and trustworthy assessments of customer valuation.

In conclusion, the review of existing literature in customer valuation underscores several pivotal research gaps, including limited industry representation beyond the FSI, the underutilization of ML in predictive analyses, and the scarcity of studies employing XAI. Addressing these gaps is essential for advancing the field, enhancing the accuracy and reliability of customer valuation methodologies, and providing actionable insights for strategic decision-making across various industry sectors.

3. Methodology

3.1. Data

The dataset utilized in this study comprises real-world transactional data obtained from a telecommunications company based in Indonesia, focusing specifically on prepaid (noncontractual) B2C customers. Prepaid customers were selected due to their dominance in the Indonesian telco market, constituting 95% of the company's total customer base [43]. The dataset covers comprehensive monthly transaction records of 200,000 clients over three years, from January 2020 to December 2022. To ensure robust analysis, only active customers in January 2020 with a minimum three-month length of stay (LOS) were included, excluding those with less than a three-month LOS to mitigate new or unstable customer biases [5]. Additionally, only B2C prepaid customers were selected to ensure individual voluntary engagement without contractual obligations. The dataset encompasses 50% active customers and 50% churned customers within the last three months of the observation period to capture diverse consumer behaviors.

Demographic data, including age, gender, and location, were incorporated into the dataset while minimizing biases. Age distribution (Fig. 1) aligns with Indonesia's demographic structure, dominated by adults (ages 25–64), seniors (age > 64), and

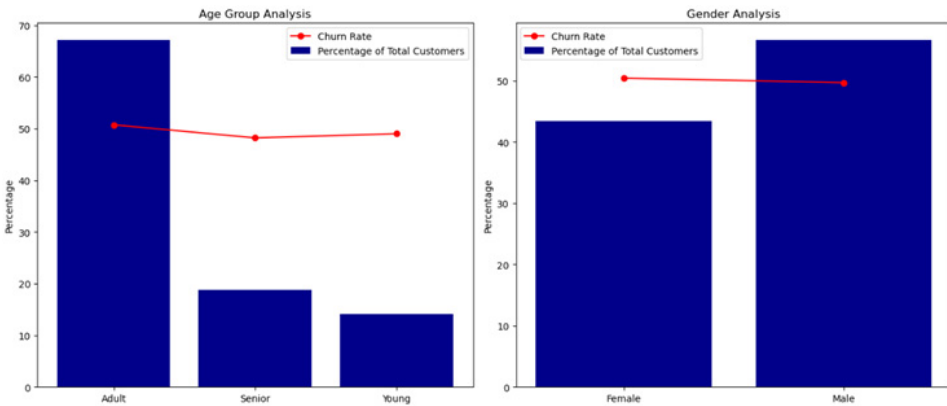


Fig. 1. Age group and gender distribution available in the experimental dataset.

young individuals (age < 25). Gender representation (Fig. 1) is balanced, and geographical regions are well represented, with slightly elevated churn rates in capital cities reflecting urban customer behavior rather than dataset biases.

The dataset provides granular revenue and usage data for various services offered by the company, including voice, SMS, broadband, digital, and international roaming. Usage metrics such as total days, transactions, duration, volume, and on-line behavior intensity are captured and aggregated monthly. Demographic details, including age, gender, and location, along with device information, offer insights into customer behavior and churn probability. A total of 141 features provide a comprehensive understanding of customer behavior and financial contributions, some of these features descriptive statistics are shown in Table 1, and the data dictionary of features is presented in Table A.1. It is also important to highlight that to ensure privacy and adhere to research ethics, customer identifiers are hashed, preserving anonymity and confidentiality. This approach underscores the commitment to ethical data analysis practices and customer privacy protection.

3.2. Modelling risk-adjusted to the revenue

This study applies the risk-adjusted valuation methodology introduced by Berger and Nasr [4], which builds on the foundational concept of CLV by incorporating a time-discounted view of future customer-generated cash flows. As discussed in Sec. 1, CLV typically accounts for both revenue and servicing costs; however, due to the unavailability of detailed cost data, only customer revenue is utilized, resulting in the computation of RAR instead of RALTV. To make the discounting process more reflective of customer-specific uncertainty — particularly relevant in prepaid telecommunications where churn and revenue fluctuations are common — this study adjusts the discount rate based on two key risk factors: the churn probability, representing the likelihood that a customer will discontinue service, and the beta value, capturing revenue volatility over time. These adaptations allow the RAR metric to reflect not only the expected financial contribution of a customer, but also the confidence or risk associated with that contribution.

Table 1. Descriptive statistics of sample features.

Metric	# Examples	Total revenue	Avg 3m	Total recharge	Avg 3m std	Total revenue std	Total charges std
mean	84	79,499	79,410	78,081	32,899	47,488	52,597
std	249	121,963	121,079	132,729	79,412	103,507	106,853
min	—	—	—	—	—	—	—
25%	9	26,490	26,447	25,000	12,917	20,651	22,664
50%	35	57,067	56,978	54,000	23,369	36,172	39,322
75%	101	104,840	104,417	100,000	39,796	58,333	62,641
max	41,299	20,319,820	20,071,050	17,581,750	16,266,500	22,733,530	20,412,410

Churn probability, a prevalent risk in the telecommunications industry especially for prepaid B2C customers, reflects the likelihood that a customer will stop using the service. Customers with a higher probability of churn pose a greater risk to future revenue, and therefore, their projected value should be discounted more heavily. To account for this, churn probability is incorporated into the discount rate following the approach proposed by Ryals and Knox [35]. This involves using the individual customer's churn probability (C_c) divided by the expected market churn probability (C_m) in the discount rate formula.^a The expected market churn probability is estimated as the average churn probability of all customers in the dataset. This method ensures that customers with above-average churn risk are assigned a higher discount rate, thereby lowering their RAR to reflect uncertainty in their future contributions. This approach, labeled "Approach A", considers three different Weighted Average Cost of Capital (WACC) values: 6.58%, 7.8%, and 7.92%, representing the minimum, median, and maximum observed WACC values during the dataset period (2020–2022).

Revenue volatility is quantified using the beta value, following the Capital Asset Pricing Model (CAPM) theory adopted by previous studies. Beta captures the extent to which a customer's revenue fluctuates to the overall customer base. A customer with a highly unstable spending pattern (high beta) represents greater financial risk. To account for this, the discount rate is adjusted based on the covariance and variance of customer revenue with market revenue.^b Here, the market represents the entire customer base of the company. Monthly assessment of customer revenue volatility enables the consideration of revenue fluctuations, enhancing the sensitivity of the RAR calculation to changes in revenue patterns. This approach, termed "Approach B", allows us to penalize customers whose revenue is more erratic, while rewarding those with stable contributions, hence offers a comprehensive evaluation of financial risks' impact on customer revenue, facilitating informed decision-making and resource allocation.

3.3. Experimental setup & ML method

The experimental design of this study involves selecting experiments to estimate the RAR value for each customer in the dataset and developing an ML model for prediction. The process begins by extracting revenue-related variables and key risk indicators — such as churn probability and revenue volatility — which are then used to compute RAR based on the previously described discounting approaches. This method provides a comprehensive framework for assessing the financial impact of different risks on customer revenue. Subsequently, rigorous statistical tests, including analysis of variance (ANOVA), are applied to quantitatively assess and evaluate the effectiveness of the proposed models, ensuring their statistical significance and potential for further investigation.

^a The discounting rate (d) is computed as $d = \frac{WACC \times C_c}{C_m}$.

^b The discounting rate d is calculated as $d = \psi_m \times \frac{cov(\varphi_{ct}, \varphi_{mt})}{var(\varphi_{mt})}$.

The RAR obtained serves as the target value for developing the ML model. Python is primarily used for ML model development, following a structured ML pipeline depicted in Fig. 2. This pipeline adheres to the Cross-Industry Standard Process for Data Mining (CRISP-DM) methodology [48], encompassing key steps such as data preprocessing, feature selection/engineering, model training, and evaluation. Each step contributes to creating a robust and accurate ML model for RAR prediction. By following this systematic approach, the model development process ensures reliability and effectiveness in predicting RAR [10].

3.3.1. Data preprocessing

A comprehensive data pre-processing strategy was employed to ensure the integrity and reliability of subsequent analyses, tailored to the specific attributes of each variable.

- **Initial Feature Selection and Data Treatment [20]:** The selection of main features involved excluding irrelevant or overly specific data points, such as postpaid customer data, and simplifying certain datasets to reduce dimensionality without compromising study objectives. Inconsistencies in data format and column names were standardized to ensure uniformity.
- **Missing Value Treatment [22]:** Missing values were handled differently based on variable characteristics. For revenue, usage, or recharge-related variables, missing values were replaced with zeroes, considering them as absent data during the observation period. Other variables followed specific rules for missing value treatment, while categorical variables either created a new category or used the mode for imputation.
- **Outlier Detection and Treatment [40]:** Outliers were detected using Z-score analysis and managed through capping, limiting extreme values. Alternative methods like Winsorizing and Binning were explored but yielded similar or lower results compared to capping, making it the preferred approach. Rows with RAR values identified as outliers were excluded from analysis to enhance data reliability.
- **Category Simplification and Encoding Categorical Variables [29]:** Sparse categories were grouped into a single category labeled “others” to reduce noise and

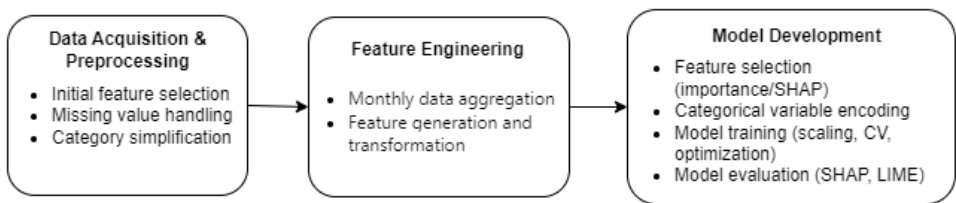


Fig. 2. Schematic framework of this study.

overfitting, ensuring a more manageable and robust dataset. Categorical variables were transformed using encoding techniques such as Ordinal Encoding for binary outcomes like gender and One-Hot Encoding for variables with more than two values. This encoding facilitated effective interpretation by ML algorithms, enhancing analysis accuracy and robustness.

3.3.2. Feature engineering

Feature engineering is a critical step in ML model development, encompassing two primary processes: feature extraction and feature selection [44]. Feature extraction involves creating new variables to offer additional insights, while feature selection identifies the most relevant features to improve the model's predictive accuracy. Several feature extraction techniques were employed in this study to enhance the dataset's interpretability and usefulness. For instance, variables such as age and region were binned to simplify data representation, allowing for more nuanced analysis. Days since the last transaction was calculated to transform datetime data into an interpretable integer format, while remaining active period was derived by subtracting the recording date from the grace period. To better capture customer behavior, service usage metrics, including the frequency of purchasing voice, broadband, or SMS packages, were introduced to assess price sensitivity. Additionally, changes in recharge and revenue were computed through percentage changes and standard deviations to measure revenue volatility. Finally, the revenue contribution of various components was calculated to gain deeper insights into customer spending patterns.

In the feature selection process [30, 44], dimensionality reduction techniques were used to simplify high-dimensional data, such as sub-district, device, and application data, thereby improving model interpretability and reducing complexity. A multicollinearity analysis was also conducted, eliminating features with correlations above 85% to avoid redundancy. Correlation analysis was employed to select the top 150 features that had the highest correlation with the target variable, ensuring the inclusion of a broad range of potential influencers. Feature importance analysis, using either embedded ML functions or global SHAP values, further refined the dataset to include only the most impactful variables. Feature scaling was applied to normalize the value ranges of the selected features, with min-max scaling chosen to standardize values between zero and one. To avoid data leakage and enhance model generalization, this scaling was applied within a pipeline framework to ensure it only affected the training data.

3.3.3. Model development, evaluation, and validation

This section outlines the construction of two ML models: the customer churn prediction model and the RAR prediction model. The churn model is used as an input in the RAR calculation, while the RAR model aims to estimate the long-term value of each customer by learning from both behavioral and financial indicators. Four

algorithms are implemented and compared: LR, valued for its simplicity and interpretability; RF and XGBoost, known for high performance in structured data environments; and CatBoost, which is particularly effective in handling categorical features without extensive preprocessing. Stratified sampling five-fold cross-validation (CV) ensured reliable model evaluation by preserving class distribution within each fold, mitigating biased evaluation risks [18]. Temporal order was disregarded during data splitting, leveraging a larger dataset for training and evaluation, enhancing robustness. The model development process involved training the base model on 150 selected features, refined using embedded ML function or SHAP values for final feature inclusion. While Principal Component Analysis (PCA) was considered for churn prediction, performance tests indicated that models trained on features selected through importance-based methods outperformed PCA in both accuracy and interpretability. For RAR prediction, PCA is excluded to maintain interpretability, which is essential in financial decision-making contexts. The optimal number of features used in each model is determined through iterative experimentation and performance evaluation.

Hyperparameter optimization utilized random search for initial exploration and grid search for fine-tuning [21], balancing computational efficiency and optimization performance. CPU and GPU resources were leveraged for LR, RF, XGB, and CatBoost algorithms, ensuring efficient exploration and optimization. Table B.1 presents the complete final set of hyperparameters used in this study.

The ML models developed in this thesis fall into two categories: the churn prediction model (a classification model) and the RAR prediction model (a regression model), each requiring specific evaluation metrics [11, 45]. For the churn prediction model, accuracy and $F1$ -score are primary metrics, with precision and recall also considered important. Accuracy measures the proportion of correctly predicted observations across all classes:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}.$$

Precision quantifies the proportion of true positive predictions among all instances classified as positive:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}.$$

Recall, also referred to as sensitivity, reflects the proportion of actual positives correctly identified by the model:

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}.$$

The $F1$ -score is the harmonic mean of precision and recall, offering a balanced measure of model performance when the dataset is imbalanced:

$$F1 - \text{score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}.$$

For the RAR prediction model, Mean Absolute Percentage Error (MAPE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2) are used as key evaluation metrics. MAPE provides an intuitive measure of prediction accuracy by expressing the average error as a percentage of the actual values, facilitating interpretability in business contexts:

$$\text{MAPE} = \frac{1}{n} \sum_{i=1}^n \left| \frac{y_i - \hat{y}_i}{y_i} \right| \times 100\%.$$

RMSE penalizes larger errors more heavily than MAPE, making it particularly suitable for applications where extreme prediction deviations are costly:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}.$$

The coefficient of determination (R^2) measures the proportion of variance in the actual values that is explained by the model. A higher R^2 value indicates a better overall model fit:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}.$$

Model robustness was assessed using various splitting strategies and CV methods [3, 26], including 5-fold CV and group-split CV, alongside learning curve analysis. XAI techniques, such as Feature Importance, SHAP, and LIME, are applied to enhance the interpretability and support *post-hoc* model validation. Feature importance provides a global overview of the most influential input variables. SHAP offers consistent, model-agnostic explanations by attributing individual prediction outcomes to feature contributions based on cooperative game theory [23]. LIME generates local surrogate models that approximate the decision boundary around a specific prediction, making it possible to understand model behavior on a case-by-case basis [32]. These methods offer insights into feature contributions globally and locally, aiding in model interpretation and validation from both performance and theoretical standpoints.

4. Results and Discussions

4.1. Probability of customer churn model

In Approach A of this study, the churn prediction model needs to be developed first, as we use the probability of customer churn as a key factor in the RAR assessment. For that, we exploit the use of LR, RF, XGB, and CatBoost algorithms. The evaluation of model performance across different numbers of features reveals that selecting 18 features yields optimal results (Fig. 3).

The performance metrics for each model are presented in Table C.1. The CatBoost algorithm outperforms logistic regression, while XGBoost shows the best

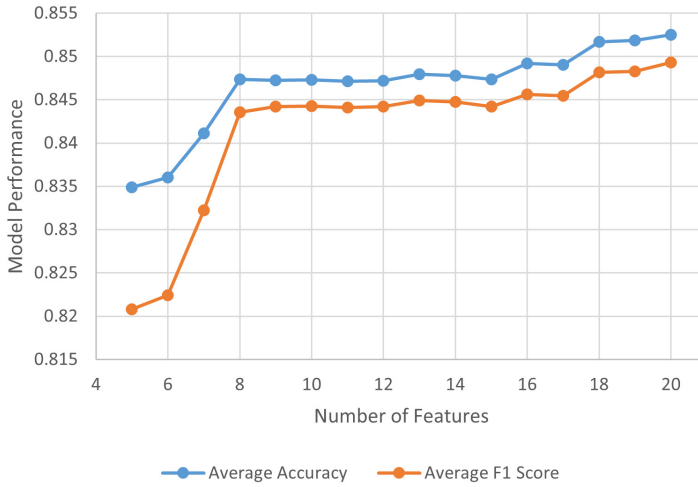


Fig. 3. Total number of features versus performance (probability of churn model, 5-fold CV).

performance, with metrics close to 0.9. The XGBoost model consistently demonstrates superior accuracy, recall, precision, $F1$ -score, and ROC-AUC. Among the evaluated models, XGBoost consistently performs best, achieving an accuracy and $F1$ -score of 0.892. Therefore, the XGBoost model is selected as the best model for this study. Detailed results, including the confusion matrix and ROC-AUC curve, are provided in Fig. D.1. Feature importance analysis (Fig. D.2) using SHAP values reveals that the type of SIM card brand used by the customer (“brand_A”) is the most important feature in predicting customer churn. Other significant features include the standard deviation of recharge transactions, SMS revenue, and various region-related features. By leveraging XGBoost and conducting feature importance analysis, we can then point valuable insights into the factors driving customer churn, facilitating informed decision-making and strategy development to mitigate churn risk.

4.2. Labeling RAR metrics

4.2.1. Approach A

Approach A for estimating RAR uses the churn probability model and the observation period’s WACC values. Minimum, median, and maximum WACC values are examined (Approaches A.1–A.3). Table D.1 shows the statistical distribution of real total revenue and computed RAR values, and Fig. D.3 shows the histogram. Customers with outlier RAR values were eliminated to establish a reliable distribution, leaving 197,692 customers. As expected, Table D.1 demonstrated that RAR values are consistently lower than total revenue. As WACC rises, RAR falls. Because the discount rate used to calculate RAR is significantly associated with WACC, this

relationship is expected. Thus, greater WACC values increase discount rates and lower RAR values.

Figure 4 shows a scatter plot of total revenue (x -axis) and calculated RAR value (y -axis), with each dot representing a client. Each dot's color indicates churn probability. The plot shows lower RAR values for increased churn likelihood. This observation supports the RAR/CLV theory, showing that the WACC and churn probability affect RAR computation and customer revenue valuation. The scatter plot shows that high-churn clients are scattered across a wide range of total revenue values. This is seen in the box plot (Fig. 5), which shows RAR values across chance of churn bins. A similar distribution of RAR values in each bin suggests that turnover likelihood affects RAR consistently across client categories. Furthermore, Fig. 6 shows that customer churn likelihood affects RAR value significantly. RAR/total income rises when churn lowers. These visuals show how total revenue, probability of turnover, and RAR affect client revenue assessment.

Finally, we should also highlight that Table D.1 and Fig. D.3 show that three WACC levels yield identical RAR values. There must be a statistical test to see if the three Approach A approaches calculate RAR values differently. The ANOVA is suitable for group mean comparisons. The data do not meet the homoscedasticity

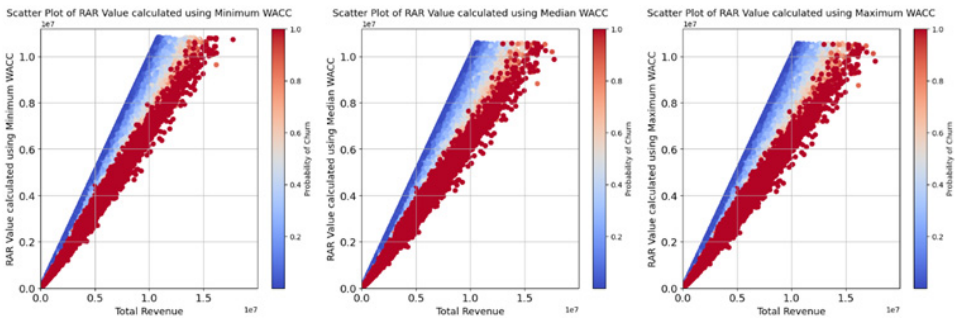


Fig. 4. Approach A — Scatter plot of total revenue versus RAR value.

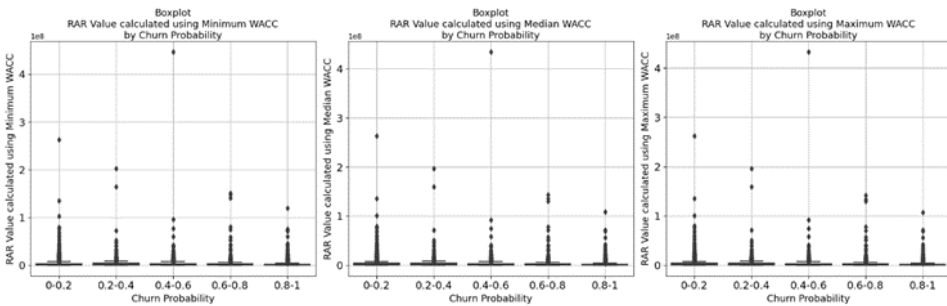


Fig. 5. Approach A — Boxplot of RAR value to probability of churn.

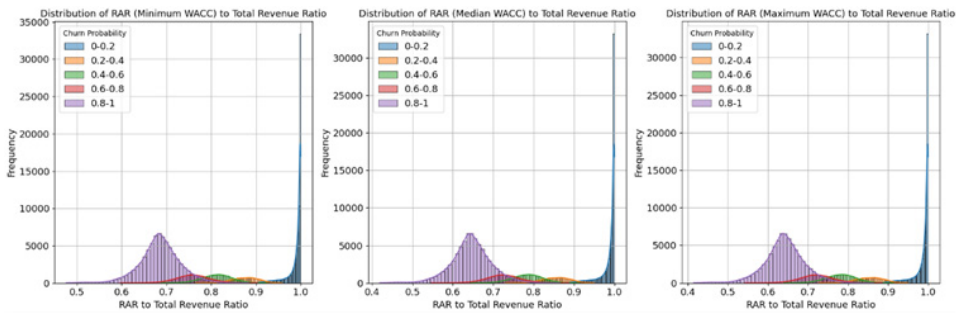


Fig. 6. Approach A — Distribution of RAR value to total revenue ratio and the probability of churn.

(equal variances) assumption for classical ANOVA. Levene's and Bartlett's tests assessed variance homogeneity. Both tests had p -values of $9.36\text{e-}08$ and $1.99\text{e-}41$, which were considerably lower than 0.05 . It suggests that data variances are not equal. Due to heteroscedasticity, Welch's ANOVA was performed. Welch's ANOVA is better for our data because it doesn't assume equal variances. Welch's ANOVA had a p -value of $2.85\text{e-}21$, well below 0.05 . This shows that technique A's three RAR methods differ statistically.

4.2.2. Approach B

Approach B results explain how RAR is calculated based on revenue volatility (Beta Value). Table D.2 shows the statistical distribution of RAR values, showing a substantial divergence from Approach A. Approach B allows larger RAR values than total revenue. This is expected because the discount rate utilized in this strategy can be positive or negative. As shown in Sec. 3.2, literature suggests to calculate the discount rate by multiplying the projected return of all corporate customers by beta. In this study, the expected return from all firm customers is -0.013 , so negative beta value leads to positive discount rate and positive beta value leads to negative discount rate and greater RAR value than actual total revenue.

Figure 7 shows the beta value distribution, which ranges from -42 to 10 . However, the scatter plot shows that few consumers have beta values below -10 and that most have beta values between -10 and 5 . The histogram figure shows that greater beta values increase RAR-to-total-revenue ratios. The boxplot also shows that beta values increase computed RAR values.

4.3. RAR predictive model: Approach A.1 (minimum WACC)

Figure 8 demonstrates MAPE and RMSE model performance for each feature count. Figure 8 shows that as the number of features increased from 5 to 9 , the average MAPE performed unstably with modest swings. The average MAPE improved consistently from nine characteristics onward, demonstrating improved prediction accuracy. Significantly, the improvement improved from 9 to 17 , hitting its lowest

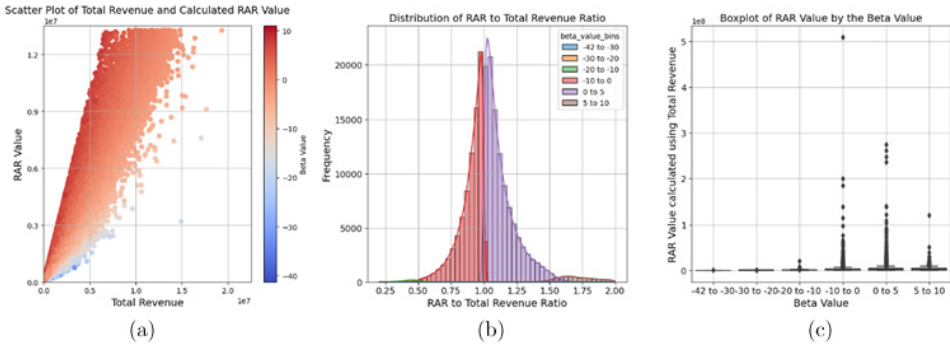


Fig. 7. Approach B — Details on (a) scatter plot, (b) histogram, and (c) boxplot of RAR value (total revenue) and beta value.

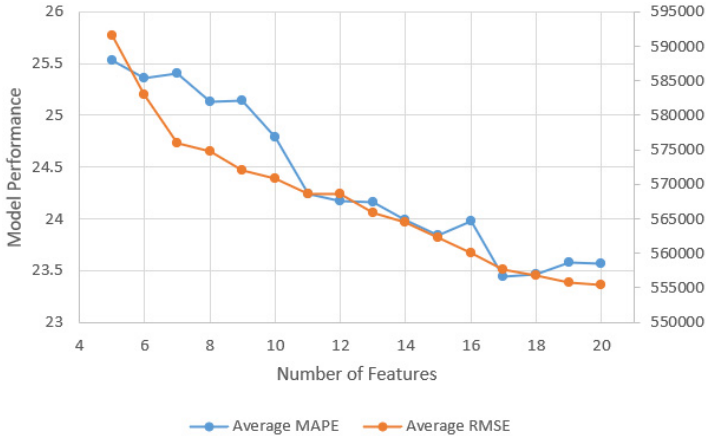


Fig. 8. Approach A.1 — Number of feature versus model performance (5-fold CV).

point at 23.45% for 17 features. In contrast, the average RMSE improved with feature count. This shows that adding features to the model improved data fit and reduced prediction error. The average RMSE dropped from 591,493 for five features to 557,624 for 17.

The best balance between prediction accuracy and model complexity was achieved with 17 features. Beyond 17 features, improvements in RMSE were marginal, and no significant reduction in MAPE was observed. The selected feature set includes a combination of variables capturing revenue volume for each services, behavioral regularity (e.g., recharge denomination, duration of activity), and reward engagement (e.g., loyalty points, loyalty tiers). While the absolute gains in RMSE beyond nine features were modest, the inclusion of additional features enhanced model robustness, generalization, and explainability. Therefore, the decision to adopt 17 features was informed by both empirical performance and the interpretability of the selected variables.

4.3.1. Predictive performance, model’s explainability and validation

The performance metrics for each model under Approach A are presented in Table 2. Among the evaluated algorithms, CatBoost consistently demonstrated superior performance. The final model achieved a MAPE of 19.38% and an R^2 value of 0.91, showing a 4% improvement in MAPE over the initial configuration while maintaining high explanatory power. The final RMSE of 558,315 is approximately 25% of the average RAR value under minimum WACC (Table D.1), indicating a reasonably accurate model. XGBoost performed comparably in early iterations, with an initial MAPE of 23.74% and R^2 of 0.91. Model refinement led to minor but consistent improvements across all metrics, including a reduced RMSE and a final MAPE of 21.94%. Despite its strong performance, XGBoost slightly lagged behind CatBoost in terms of final accuracy. RF showed the weakest improvement trajectory. Although it started with reasonable performance, the final MAPE remained above 22% and RMSE reductions were minimal. R^2 stayed relatively constant at 0.90 throughout the development process. Overall, CatBoost was selected as the final model due to its balanced and superior performance across all key evaluation metrics.

Global Explanation: Using the model’s feature importance function and SHAP values’ global interpretation, feature importance analysis was performed to validate the model. Figure D.4 shows that both strategies yielded the top nine most essential traits. The feature importance technique found that “poin”, the customer’s total loyalty points from past transactions, was the most influential characteristic. This attribute had the most importance and affected model predictions. “Poin” was twice as important as the second most important factor, churn likelihood. When considering SHAP global relevance, the “poin” attribute had less impact on prediction outcomes. SHAP values showed how feature values affect anticipated RAR values.

Table 2. Approach A.1 — Model performance metrics (5-fold CV).

Model and model development stage	MAPE	MAE	RMSE	R^2
CatBoost				
Initial model and all features	23.36	358,254	548,076	0.91
Feature selection	23.64	362,555	555,542	0.91
Random search hyperparameter tuning	19.51	355,476	560,046	0.91
Grid search hyperparameter tuning	19.38	353,781	558,315	0.91
XGBoost				
Initial model and all features	23.74	369,528	565,848	0.91
Feature selection	23.66	371,903	569,962	0.91
Random search hyperparameter tuning	22.45	363,472	559,686	0.91
Grid search hyperparameter tuning	21.94	360,894	557,307	0.91
Random Forest				
Initial model and all features	23.13	379,640	584,747	0.9
Feature selection	23.09	381,036	585,703	0.9
Random search hyperparameter tuning	22.58	379,528	584,432	0.9
Grid search hyperparameter tuning	22.56	378,131	582,710	0.9

SHAP values showed that greater “poin” values were connected with higher RAR values, demonstrating a positive association between customer loyalty points and revenue.

Churn probability was the second most important factor in feature significance and SHAP analysis. This underscores its importance in churn prediction. The third most important parameter was the average revenue over the last three months (“avg_3m”), demonstrating the importance of recent revenue patterns in understanding customer attrition.

Local Explanation: More comprehensive research utilizing SHAP waterfall plots and LIME examined how each feature affected the prediction result. These approaches reveal the contribution of each feature for particular cases, enabling a more detailed understanding of their prediction impact. Figure D.5 shows the results of this analysis of sample X, which represents stayed customers. For the selected instances, SHAP and LIME identified customer churn likelihood as the most relevant factor. An increase in turnover probability improved the expected result in both models. Similar characteristics include “hvc_tier_gold”, “avg_3m_pct_change” and “poin” in both models’ top eight most influential features. Notably, certain features had distinct influence orientations between the two models. SHAP benefited from the “hvc_tier_gold” feature, whereas LIME suffered.

Comparison with sample Y provided a broader view. As sample Y has a high churn rate, the study sought to identify feature importance patterns like Fig. D.6. In contrast to sample X, the top five SHAP-based traits contributed negatively to the anticipated value. The most important feature, “poin”, contributed 377,204 to the expected result, followed by “prob_churn” with 326,204. The varying values and interactions of each feature may cause this difference. LIME found that “hvc_tier_Gold”, which contributed 366,469 to sample Y’s projected RAR value, was the most important characteristic. “avg_3m_pct_change” contributed 250,855.

Dependence Plots: To determine feature-SHAP value interactions, SHAP dependence plot analysis was performed. Prob_churn was the major feature and the other top five features — poin, “avg_3m_std”, “avg_3m”, and “total_revenue” — were subsidiary features. Figure 9 shows dependent charts showing how SHAP value affects customer churn likelihood. Churn probability and SHAP are negatively correlated. As churn probability grows, SHAP value contribution to anticipated result diminishes. The association between churn likelihood and RAR from the theory supports this result.

The “prob_churn” value and other features have different effects on SHAP results. Several “prob_churn” values could contribute to the same SHAP value. The interaction with another feature was examined to understand this occurrence. The “poin” feature interaction shows that low probability of churn with high “poin” value has better SHAP value contribution than low probability with low “poin”. SHAP value contribution drops with high churn probability and “poin” value. Other selected secondary characteristics “avg_3m_std”, “avg_3m”, and “total_revenue” also show

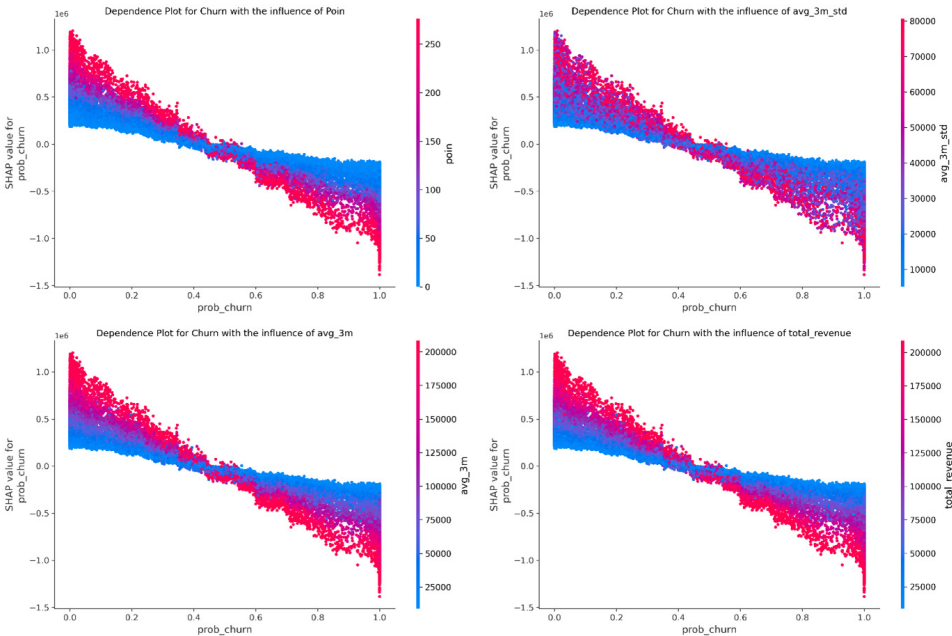


Fig. 9. Approach A.1 — SHAP dependence plots of probability of churn.

same pattern in their dependent plots. It makes sense as the “poin” element indicates customer spending history, and the other features might represent customer expenditure as well. Thus, all identified secondary factors interact similarly with churn likelihood.

Table 3 shows the results of several splitting strategies used to test the final model’s robustness. The table shows that performance measures vary slightly when the split ratio varies from 50:50 to 90:10, but the model’s performance remains steady. Interestingly, all metrics perform similarly with almost no changes. The 80:20 ratio had the best MAPE and MAE, whereas the 70:30 ratio had the best RMSE. Furthermore, all splitting algorithms maintained strong R^2 values of 0.91. These results show that the model is resilient and stable across split ratios, proving its capacity to generalize to new data.

Table 3. Approach A.1 — Final model performances across different split strategy.

Split ratio	MAPE	MAE	RMSE	R^2
50:50	19.62	356,987	561,280	0.91
60:40	19.50	355,546	560,821	0.91
70:30	19.32	353,537	559,225	0.91
80:20	19.27	352,344	561,539	0.91
90:10	19.36	354,396	562,349	0.91

The final model was further analyzed using 5-fold CV to provide a complete understanding of model validation and ensure that it is not overfitted and stable. Below is Table 4 with the results. The training and test sets perform similarly across all assessment criteria throughout the five folds. This shows that the final model is not overfitting because it performs well on unseen data and training data. Performance measures do not vary much over folds, demonstrating model stability. This shows that the model's performance is not substantially dependent on any data subset, proving its generalizability and resilience. The group split approach was used to test the model's performance under multiple splitting strategies, as shown in Table 5. Traditional CV algorithms predicted that the model's training set performance metrics would be better than its test set metrics. The test set results fluctuated slightly but not significantly. The best MAPE on the test set was 18.94% for test_group 4. The worst MAPE was 21.38% in test_group 3. These changes may be due to regional data differences. Region tier 3, the middle-low region, may have patterns that the model trained on data from earlier regions cannot capture.

The learning curves were also extracted and analyzed to explore model's learning efficiency and robustness based on the number of training data used. We could observe that the training scores increase as the training set grows, but Validation_scores fall. The increasing training error shows the model's decreasing ability to match training data correctly, but the lowering validation error shows its improving generalization to unseen data. Training and validation scores are close, especially

Table 4. Approach A.1 — Final model 5-fold CV result.

Fold	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	15.87	290,105	474,393	0.93	19.38	352,691	560,862	0.91
2	15.94	290,944	476,666	0.93	19.28	352,339	552,498	0.91
3	15.74	289,685	476,449	0.93	19.30	354,723	556,717	0.91
4	15.88	288,910	472,150	0.94	19.39	357,390	567,578	0.91
5	15.88	290,527	475,742	0.93	19.54	351,761	553,919	0.91
Avg	15.86	290,034	475,080	0.93	19.38	353,781	558,315	0.91

Table 5. Approach A.1 — Final model group-split result.

Test group	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	15.84	284,610	467,069	0.94	19.76	402,326	622,598	0.9
2	15.48	289,085	475,847	0.94	19.09	338,876	536,929	0.91
3	15.28	289,954	474,848	0.94	21.38	341,684	541,700	0.91
4	16.17	286,981	467,675	0.93	18.94	518,462	755,218	0.88
Avg	15.69	287,658	471,360	0.94	19.79	400,337	614,111	0.90

with higher training numbers. This suggests the model isn't overfitting because it generalizes effectively to new data.

The ML model's impacting features were compared to traditional calculation methods' variables to validate the model's conclusions. Comparisons improve the model's coherence and capacity to identify key traits. Approach A's RAR computation consider the discount rate from the WACC value multiplied by the customer churn probability and divided by the market's expected probability. Thus, traditional RAR calculation relies on customer churn likelihood. The resulting model also ranks churn probability second in feature importance and SHAP global explanations. The local explanation from two randomly selected data samples likewise revealed churn probability as the most critical factor for SHAP and LIME. Figure 9's dependent plot shows a negative link between churn likelihood and SHAP value. As churn likelihood rises, SHAP value contribution to anticipated RAR declines or becomes negative. According to the classic RAR calculation approach, a higher churn probability leads to a higher discount rate and a lower RAR value.

The final model's other most essential features ("point", "avg_3m_std", "total_revenue", and "total_revenue_std") improve model coherence with standard calculation methods. These attributes represent client expenditure or revenue. This is also in accordance with the literature, which calculates RAR by summing the customer's revenue over the observation time and dividing by the discount rate raised to the power of the duration. Two of these features, "avg_3m_std" and "total_revenue_std", show client spending volatility, which is expected given six semesters of observations. Volatility affects RAR revenue, hence the final model prioritizes these attributes.

4.4. RAR predictive model: Approach A.2 (median WACC)

Figure 10 shows the relationship between Approach A.2's feature selection performance and feature number. Similar to the previous strategy, the average MAPE fluctuates from 5 to 9 characteristics before improving to continuously improve. The MAPE achieved 23.85% with 18 characteristics, then improved slowly. In contrast, the average RMSE improves with more characteristics. This shows that adding features improves model fit to observed data, reducing prediction errors. The average RMSE drops from 587,396 for five features to 545,709 for 18.

Based on this trend, 18 features were selected for the final model to balance predictive accuracy and model complexity. While further feature additions might offer incremental gains, the observed performance plateau beyond 18 features suggests limited added value. The selected features include variables related to revenue magnitude and stability, user engagement (e.g., recharge and data usage frequency), and loyalty tier. This combination supports both the predictive power and the interpretability of the model, making it a practical configuration for real-world implementation.

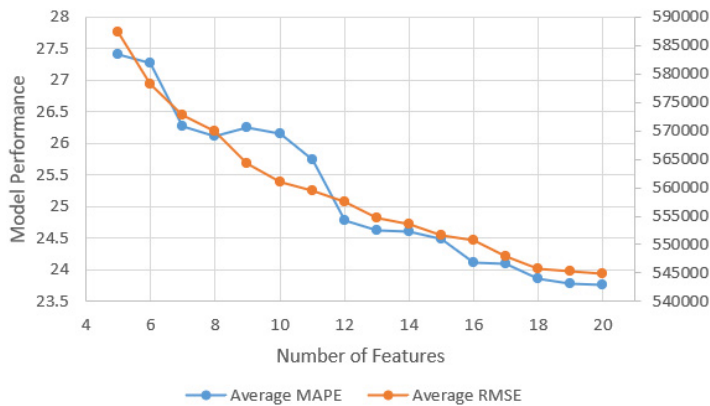


Fig. 10. Approach A.2 — Number of feature versus model performance (5-fold CV).

4.4.1. Predictive performance, model's explainability and validation

Table 6 shows model performance indicators at various stages of development. The CatBoost algorithm excels in all measures. After feature selection and hyperparameter tuning, the final CatBoost model achieved the lowest MAPE of 19.61%, the lowest MAE of 346,318, and maintained a high R^2 of 0.91. These results reflect both high predictive accuracy and model stability. As in Approach A.1, the RMSE of 547,301 corresponds to approximately 25% of the average RAR value, providing further context for evaluating the model's error margin.

XGBoost showed modest improvement during model refinement, ending with a MAPE of 22.21% and a similar R^2 of 0.91. RF demonstrated the weakest performance, with only marginal gains throughout development and final metrics trailing

Table 6. Approach A.2 — Model development (5-fold CV).

Model and model development stage	MAPE	MAE	RMSE	R^2
CatBoost				
Initial model and all features	23.21	349,567	535,188	0.91
Feature selection	23.85	355,790	545,710	0.91
Random search hyperparameter tuning	20.33	353,120	556,118	0.90
Grid search hyperparameter tuning	19.61	346,318	547,301	0.91
XGBoost				
Initial model and all features	24.21	364,263	559,590	0.904
Feature selection	23.90	365,209	561,521	0.904
Random search hyperparameter tuning	22.91	359,090	551,957	0.902
Grid search hyperparameter tuning	22.21	353,787	548,270	0.91
Random Forest				
Initial model and all features	24.52	373,964	577,500	0.9
Feature selection	24.47	375,088	577,966	0.9
Random search hyperparameter tuning	24.44	374,299	576,814	0.9
Grid search hyperparameter tuning	23.59	372,164	574,789	0.9

behind both CatBoost and XGBoost. Given its consistently superior performance and balanced results across MAPE, RMSE, MAE, and R^2 , CatBoost was selected as the final model for this approach.

Global Explanation: The best model underwent feature important analysis using the embedded feature importance approach and SHAP explanation using SHAP summary plot. Figure D.7 showed consistent results with the previous strategy. The most essential trait, “poin”, has double the importance of “prob_churn”. Meanwhile, “avg_3m”, the next most important parameter, has less than half the impact of “prob_churn”. SHAP values confirmed feature importance findings. As in the prior experiment, “poin” was the most essential feature. The difference in feature importance is greater than the difference in SHAP value impact between the most important feature and the other top four features.

Local Explanation: Local explanations using SHAP value and LIME were used to better comprehend each feature’s impact. This analysis used the same data sample as before. Figure D.8 shows that SHAP value and LIME consistently showed that the top four traits had a similar impact direction to the previous technique, although the magnitude differed. The most relevant feature according to SHAP value and LIME was “prob_churn”, which improved the expected result. The remaining features have different rankings and influence orientations than SHAP value and LIME. LIME does not list “avg_3m”, the second most essential feature by SHAP value, in its top eight. While SHAP value gives distinct influence direction, LIME finds “avg_3m_pct_change” as the second most important feature, closely related to “avg_3m”. Despite SHAP global explanation ranking “poin” as the most significant feature, SHAP value ranks it lower and LIME for this instance does not list it among the top eight key characteristics.

The second comparison sample was data sample Y. Strangely, the top features negatively predicted SHAP values. SHAP found that “prob_churn” negatively contributed 380,500, followed by “poin” with 272,048. LIME found that “avg_3m_pct_change” was the most important feature, contributing 444,660 negatively, while “hvc_tier_Gold” contributed 412,575. Another important element was “day_of_recharge”, ranking third and contributing 399,709. These contrasts show how SHAP values and LIME analyze and value aspects for each occurrence. Figure D.9 shows the details described here.

Dependence Plot: Figure D.10 shows the dependence plot of “prob_churn” with the other top five features (“poin”, “avg_3m” and “avg_3m_std”). The chance of churn and SHAP value were negatively correlated, as in the previous approach. Instances with a low chance of churn and a high “poin” value had a larger SHAP value contribution than those with a low “poin” value. The SHAP value contribution reduced when the churn probability and “poin” value were high.

The influence of “avg_3m_std” on SHAP value and churn likelihood was less clear than the other characteristics. A high “avg_3m_std” value and low churn probability

could nonetheless result in a poor SHAP value. This happens because “avg_3m_std” is slightly different from other features. While the other attributes represent historical client spending, “avg_3m” represents the average revenue from customers in the last three months, and “avg_3m_std” measures its standard deviation. Thus, its connection with “prob_churn” is more complicated.

The best model from technique A.2 was validated using multiple splitting strategies to evaluate performance. Table 7 shows findings from 50:50 to 90:10 splitting ratios. The data show stable performance across splitting ratios, with 90:10 performing best. These results show that the model is resilient and stable across split ratios, proving its capacity to generalize to new data.

Table 8 contains a 5-fold CV process result. Like the prior method, performance consistency across all assessment measures and folds for training and test sets is evident. This consistency shows that the model handles overfitting and performs equally with unseen and training data. Minor performance metrics fluctuation over folds suggests that no single subset of the data affects the model’s performance, indicating model stability.

Table 9 shows that group split analysis produced similar results to the previous method, with modest performance differences. In particular, test_group 4 had the highest MAPE, highest RMSE, and lowest R^2 value. Similar to the prior strategy, test_group 3 had the worst MAPE. Note that test_group 4’s weaker MAE and RMSE performance may be due to its inclusion of places with lower economic power, which

Table 7. Approach A.2 — Final model performances across different split strategies.

Split ratio	MAPE	MAE	RMSE	R^2
50:50	20.14	352,299	559,715	0.91
60:40	19.75	348,603	554,279	0.91
70:30	19.7	349,510	556,280	0.91
80:20	19.86	347,267	552,860	0.91
90:10	19.64	344,555	546,625	0.91

Table 8. Approach A.2 — Final model 5-fold CV results.

Fold	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	16.3	287,640	469,430	0.93	19.77	347,029	551,947	0.91
2	16.25	289,099	472,341	0.93	19.64	346,259	550,746	0.91
3	16.36	286,834	469,160	0.93	19.5	343,313	540,605	0.91
4	16.32	286,427	468,447	0.93	19.41	346,215	543,951	0.91
5	16.33	287,466	469,619	0.93	19.73	348,773	549,256	0.91
Avg	16.31	287,493	469,799	0.93	19.61	346,318	547,301	0.91

Table 9. Approach A.2 — Final model group-split results.

Test group	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	16.29	282,609	464,647	0.93	20.1	393,836	607,354	0.9
2	15.98	286,177	469,020	0.94	19.34	332,385	526,891	0.91
3	15.7	287,718	471,364	0.93	21.62	334,331	530,774	0.91
4	16.57	284,549	463,970	0.93	19.02	508,967	744,998	0.87
Avg	16.14	285,263	467,250	0.93	20.02	392,380	602,504	0.90

may limit the model’s capacity to generalize to these regions. Despite these minor error rates, the model performs well across all regions.

Learning curves are extracted and analyzed to explore model validation and assess learning efficiency and resilience. We can observed that as the training set size increases, the curves show an increase in training scores and a drop in validation scores. This demonstrates the model’s ability to generalize unseen data as it fits training data less well. The limited gap between training and validation scores, especially at larger training sizes, suggests the model is not overfitting.

A comparison between influential ML model features and traditional calculation variables validates model outputs. Like Approach A.1, customer attrition likelihood affects traditional RAR calculation, and the ML model does too. It shows that churn likelihood is the most essential factor in both global and local explanations. The dependence plot (Fig. D.10) shows a negative association between SHAP value contribution and probability of churn, with greater probabilities resulting in lower SHAP values. The importance given to Approach A.2’s top attributes shows that it closely resembles traditional computation approaches. Like before, “poin”, “avg_3m_std”, “total_revenue”, and “total_revenue_std” are crucial. These attributes represent client expenditure and revenue segments. This interpretation is in accordance with the RAR definition and calculation, bolstering the model’s robustness. Interestingly, “avg_3m_std” and “total_revenue_std” represent consumer spending variances across six semesters. These movements affect RAR value, making them the model’s most essential aspect.

4.5. RAR predictive model: Approach A.3 (maximum WACC)

MAPE and RMSE were used to evaluate the model’s performance based on feature count in Fig. 11. Similar to the prior approach, MAPE performance fluctuated throughout the observation, while RMSE improved continuously. We can identify that 17 features is the optimal number of variables for the final model. A trade-off between model performance and complexity determined this option. The average MAPE and RMSE did not improve with more features than 17. Thus, picking 17 characteristics was best for balancing model correctness and complexity.

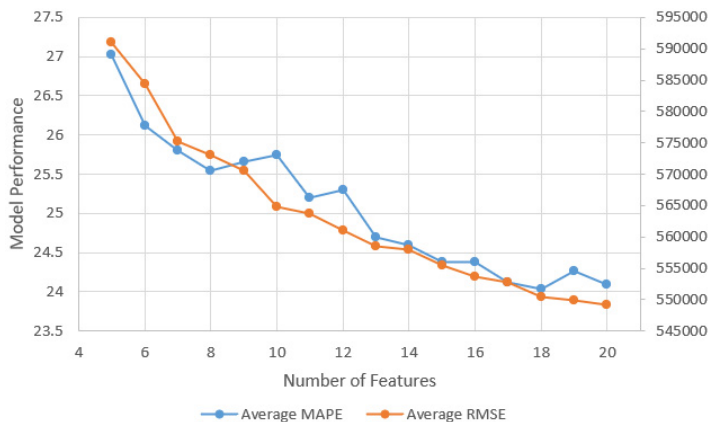


Fig. 11. Approach A.3 — Number of feature versus model performance (5-fold CV).

4.5.1. Predictive performance, model’s explainability and validation

Table 10 summarizes model performance across development stages for Approach A.3. CatBoost delivered the best results overall, with its final model achieving a MAPE of 19.80%, MAE of 348,060, RMSE of 550,109, and R^2 of 0.908. While RMSE fluctuated slightly during tuning, MAPE consistently improved from 23.86% in the initial model, and R^2 remained stable across all iterations. These results indicate strong and stable predictive performance. The final RMSE is slightly higher than 25% of the average RAR value in this approach, indicating that the prediction error remains within a reasonable range relative to the target scale.

XGBoost showed steady improvement throughout development, ultimately achieving an RMSE of 548,878 and an R^2 of 0.91, matching CatBoost. However, it

Table 10. Approach A.3 — Model development (5-fold CV).

Model and model development stage	MAPE	MAE	RMSE	R^2
CatBoost				
Initial model and all features	23.86	352,323	540,175	0.91
Feature selection	23.35	352,475	541,102	0.91
Random search hyperparameter tuning	19.92	351,129	554,663	0.906
Grid search hyperparameter tuning	19.80	348,060	550,109	0.908
XGBoost				
Initial model and all features	24.14	363,165	558,302	0.908
Feature selection	24.03	364,693	559,999	0.902
Random search hyperparameter tuning	22.75	354,497	547,482	0.91
Grid search hyperparameter tuning	22.19	353,865	548,878	0.91
Random Forest				
Initial model and all features	24.19	373,054	576,709	0.9
Feature selection	23.96	374,254	576,936	0.9
Random search hyperparameter tuning	23.55	372,967	576,047	0.9
Grid search hyperparameter tuning	23.39	371,705	574,476	0.9

lagged behind CatBoost in both MAPE (22.19% versus 19.80%) and MAE (353,865 versus 348,060), which are more indicative of prediction accuracy in this context. RF consistently underperformed the other models, with a final MAPE of 23.39% and the highest RMSE and MAE values. Given CatBoost’s superior accuracy and balanced performance across all key metrics, it was selected as the final model for this approach.

Global Explanation: As before, the top nine attributes and SHAP global explanation remain the same. Figure 12 shows that “poin” is the most influential attribute, much outranking “prob_churn”. The SHAP summary graphic confirms that “poin” affects the anticipated result, but not as much as the feature importance values suggest. “Prob_churn” and “avg_3m” remain SHAP’s second and third most essential features. This method resembles the previous one.

Local Explanation: As in earlier techniques, “prob_churn” is the most relevant feature in the local explanation employing SHAP and LIME in Fig. D.11. Other feature rankings and contribution directions change between techniques. “prob_churn” contributes 738,068 SHAP value to the expected outcome. Interestingly, LIME ranks “redeemer_mtd_flag” as the second most essential feature, with a contribution value similar to “prob_churn” (1,138,101 versus 1,096,596). SHAP value does not list “redeemer_mtd_flag” among the top nine features.

SHAP and LIME likewise ranked “prob_churn” as the most relevant attribute for sample Y (Fig. D.12). Both have distinct additional important traits. SHAP’s top three most important attributes are prob_churn, poin, and remaining active period, which contributed negative values of 344,758, 288,457, and 288,060. However, LIME

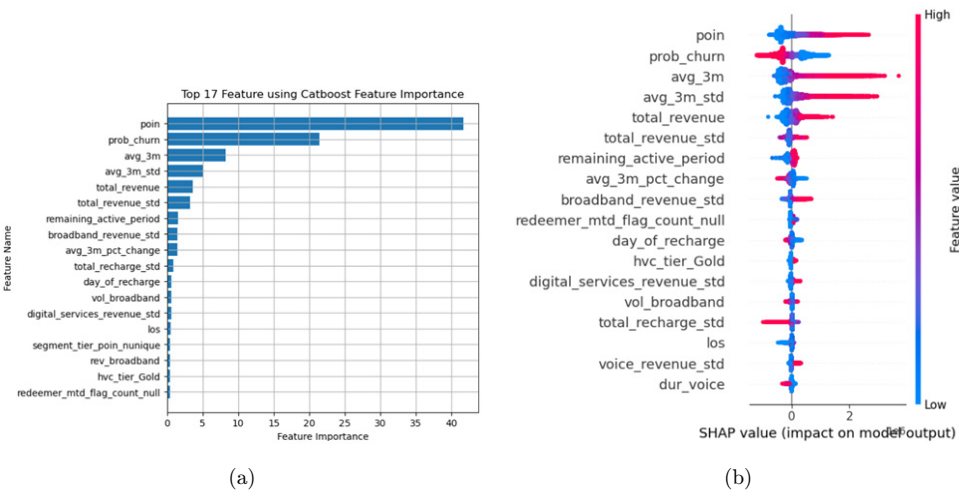


Fig. 12. Approach A.3 — Explainability with (a) feature importance and (b) SHAP value.

ranks “prob_churn”, “redeemer_mtd_flag_count_n” and “day_of_recharge” as the top three characteristics, contributing $-801,250$, $-727,918$ and $331,895$.

Dependence Plots: Figure 13’s dependence plot analysis demonstrates results consistency among techniques. Similar to prior studies, the chance of churn was negatively correlated with SHAP, indicating a continuous influence on predictions. The interaction with the other top five factors showed similar tendencies, demonstrating that these features affect SHAP values consistently across techniques. The findings are comparable since the RAR was calculated using the same approach, with only the WACC value changing. This unified technique assures that each feature’s effect on SHAP value is similar across approaches.

The best model from technique A.3 was validated using multiple splitting strategies to evaluate performance. Table 11 shows findings from 50:50 to 90:10 splitting ratios. The data shows stable performance across splitting ratios, with 90:10 performing best. These results show that the model is resilient and stable across split ratios, proving its capacity to generalize to new data.

Table 12 shows the 5-fold CV result for this strategy. Performance on all assessment criteria across all five folds for both the training and test sets is consistent, as with previous models. This means the model has mitigated overfitting and performed well on training and unseen data. The performance indicators’ tiny fold changes confirm the model’s stability, indicating that no data subset significantly affects its predictions. Group split analysis verified the model’s robustness across

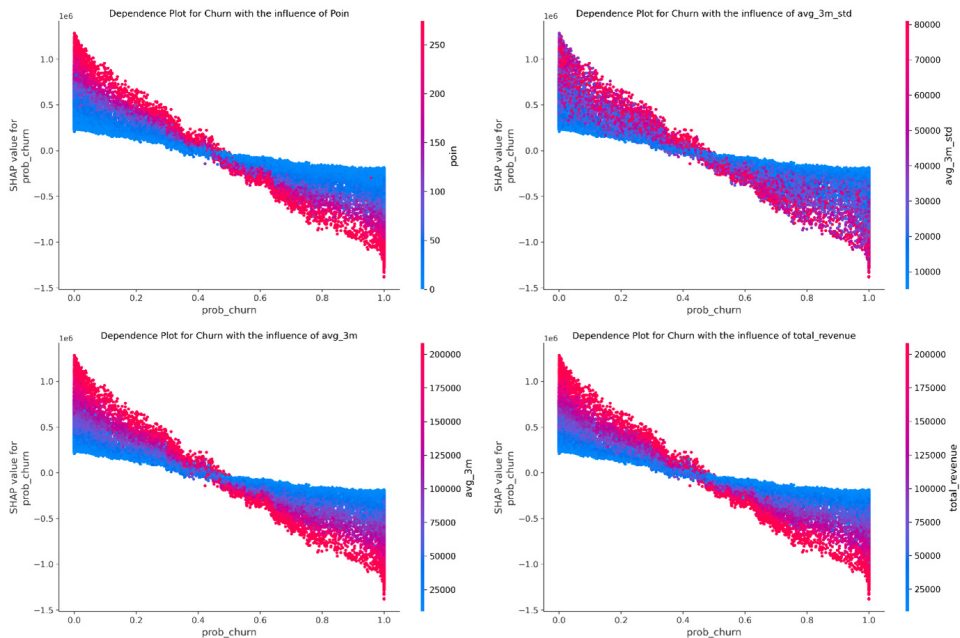


Fig. 13. Approach A.3 — SHAP dependence plot of probability of churn.

Table 11. Approach A.3 — Final model performances across different split strategies.

Split ratio	MAPE	MAE	RMSE	R^2
50:50	19.96	348,378	547,198	0.91
60:40	19.87	346,241	543,214	0.91
70:30	19.58	344,013	538,791	0.91
80:20	19.49	342,713	539,018	0.91
90:10	19.21	341,508	539,892	0.91

Table 12. Approach A.3 — Final model 5-fold CV results.

Fold	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	15.41	264,830	438,019	0.94	19.5	343,168	540,756	0.91
2	15.44	266,646	442,098	0.94	19.71	344,521	540,601	0.91
3	15.4	264,617	437,423	0.94	19.72	347,818	550,107	0.91
4	15.34	263,399	436,265	0.94	19.63	352,027	557,146	0.91
5	15.2	265,289	439,618	0.94	19.84	348,501	555,104	0.91
Avg	15.36	264,956	438,685	0.94	19.68	347,207	548,743	0.91

groups. The results are in Table 13. As with earlier approaches, performance measures fluctuated slightly, suggesting robustness across all groups. In line with prior findings, train_group 3 had the highest MAPE value at 22.15% and train_group 4 had the highest MAE and RMSE values.

Learning curves are extracted and analyzed to assess model learning efficiency and robustness. We can observe that as the training set size increases, training scores rise and validation scores fall. This shows the model’s improving ability to generalize unseen data while fitting training data less precisely. The small disparity between training and validation scores, especially at larger training sizes, supports the model’s nonoverfitting.

Validating the model’s outputs is also important, and the ML model’s significant features are compared to traditional calculation methods’ variables. As in prior

Table 13. Approach A.3 — Final model group-split results.

Test group	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	15.35	260,924	434,420	0.94	20.1	393,687	608,595	0.89
2	14.95	259,692	433,220	0.95	19.43	332,649	527,872	0.91
3	14.71	262,135	436,194	0.94	22.15	334,709	530,085	0.91
4	15.63	264,314	436,086	0.94	19.42	510,876	745,660	0.87
Avg	15.16	261,766	434,980	0.94	20.28	392,980	603,053	0.90

approaches, the probability of churn is significant in traditional RAR calculation, and the final model in Approach A.3 acknowledges its importance in global and local explanation. Probability of churn and SHAP value on dependence plot analysis (Fig. 13) both show a negative link.

By assigning relevance to characteristics like “poin”, “avg_3m_std”, “total_revenue”, and “total_revenue_std”, the ML model and traditional calculation methods become more closely linked. These attributes provide unique insights on customer spending patterns and revenues, but are still in accordance to the RAR literature. The model also prioritizes “avg_3m_std” and “total_revenue_std” to represent client spending volatility across six semesters. The model’s capacity to account for volatility in RAR calculations matches established approaches and strengthens its relevance and reliability.

4.6. RAR predictive model: Approach B (beta value)

Figure 14 gives a complete MAPE and RMSE analysis of the model across different feature numbers. As features increase, both metrics improve. Starting with five features, the MAPE drops from 25.08% to 21.51% at 17 features. Adding more features improves predicted accuracy, as shown by this significant MAPE reduction. Similarly, RMSE decreases from 709,734 to 659,678 when employing 17 features. Beyond 17 features, further reductions in RMSE are minimal, showing diminishing prediction error reduction. These findings indicate that 17 features are the best configuration for this method. The choice of 17 features balances predictive power (lower MAPE) and interpretability, supported by feature importance insights shown later in the explainability analysis.

4.6.1. Predictive performance, model’s explainability and validation

Table 14 summarizes model performance throughout development under Approach B. CatBoost consistently delivered the best results in terms of overall accuracy and

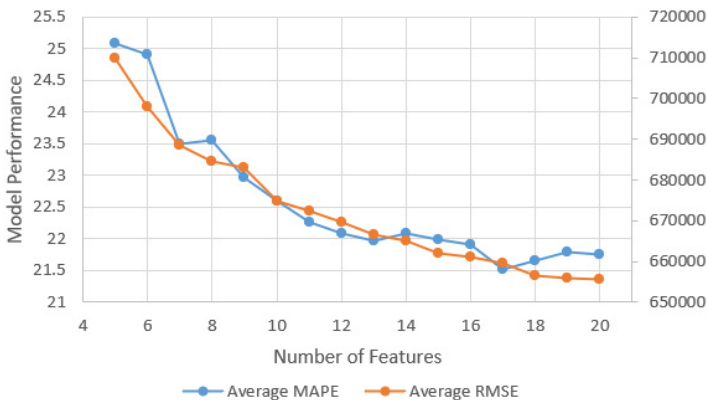


Fig. 14. Approach B — Number of feature versus model performance (5-fold CV).

Table 14. Approach B — Model development (5-fold CV).

Model and model development stage	MAPE	MAE	RMSE	R^2
CatBoost				
Initial model and all features	21.77	421,813	647,108	0.92
Feature selection	21.44	429,666	660,020	0.92
Random search hyperparameter tuning	18.67	428,995	677,393	0.91
Grid search hyperparameter tuning	18.35	422,314	665,811	0.92
XGBoost				
Initial model and all features	22.04	436,544	671,285	0.91
Feature selection	21.84	441,744	679,921	0.91
Random search hyperparameter tuning	20.32	427,497	660,792	0.92
Grid search hyperparameter tuning	19.99	427,745	664,809	0.92
Random Forest				
Initial model and all features	21.84	451,157	698,535	0.91
Feature selection	21.82	452,722	698,729	0.91
Random search hyperparameter tuning	21.05	449,444	696,134	0.91
Grid search hyperparameter tuning	21.14	449,315	695,449	0.91

stability. After hyperparameter tuning via grid search, the final model achieved a MAPE of 18.35%, MAE of 422,314, RMSE of 665,811, and R^2 of 0.92. These results reflect a notable improvement in MAPE from the initial configuration (21.77%), while R^2 remained stable. Although RMSE and MAE fluctuated slightly during tuning, the final RMSE of 665,811 corresponds to approximately 24% of the average RAR value for Approach B (Table D.2), indicating a reasonable prediction error relative to the scale of the target variable and maintained competitive values across all metrics.

XGBoost followed a similar development path, improving MAPE from 22.04% to 19.99%, and matching CatBoost in R^2 . It slightly outperformed CatBoost in final MAE and RMSE; however, its MAPE remained higher. RF showed limited improvement across development stages, ending with a final MAPE of 21.14%, the highest RMSE, and the lowest R^2 of the three models. Given CatBoost’s superior MAPE and balanced performance across all evaluation criteria, it was selected as the final model for this approach. Although XGBoost achieved marginally lower MAE and RMSE, CatBoost’s overall accuracy and consistency made it the more robust choice.

Global Explanation: This RAR calculation method, unlike others, accounts for customer revenue volatility by using beta value as a risk factor. Although this strategy differs from others, the model’s most crucial attribute, “poin”, remains consistent. With approximately four times the importance of the second most essential trait, “poin” dominates. The SHAP value analysis also ranks “poin” as the most essential trait, but its impact is unchanged. “Beta_total_return”, the beta value for the customer’s total return, is the second most essential feature by feature importance and the third by SHAP value. This strategy ranks “prob_churn” among the

top 10 most essential attributes, but its impact is lower than in the other way. The result is detailed in Fig. 15.

Local Explanation: Local explanation on the same data instance as the previous approach reveals SHAP-valued impacting aspects. Figure D.13 shows that “total_revenue” and “avg_3m” positively affect the anticipated result with equal SHAP values by 467,272 and 465,637. “beta_total_return” has a 100,394-point positive impact, but “prob_churn” is not in the top 10 for this data point. However, according to LIME, “beta_total_return”, which positively affects the anticipated value by 1,762,681, is the most relevant attribute. This value is more than double the second most important attribute, “hvc_tier_Gold”, which has a negative LIME value of 794,218. Interestingly, LIME ranks “prob_churn” among the top five feature importances with a positive contribution of 307,242.

Sample Y, with a negative beta value, was used for comparison. This sample underwent SHAP and LIME analysis, as shown in Fig. D.14. The most influential feature in SHAP and LIME explanations was “beta_total_return”. The SHAP analysis showed that the negative beta value contributed $-1,055,187$ to the projected result. LIME also showed that beta had a negative impact of 3,001,641. We should also highlight the findings in Approach B are in accordance to the RAR literature, meaning that as seen, beta and RAR are positively correlated. Sample Y has a negative beta value of -3.05 , hence its RAR value is lower than its total revenue. The expected result is negatively affected by the negative beta value.

Dependence Plot: SHAP dependence plot research examined the interaction between features and SHAP values in Approach B. “poin”, “avg_3m”, “beta_total_return”, “total_revenue”, and “avg_3m_std” were the SHAP global

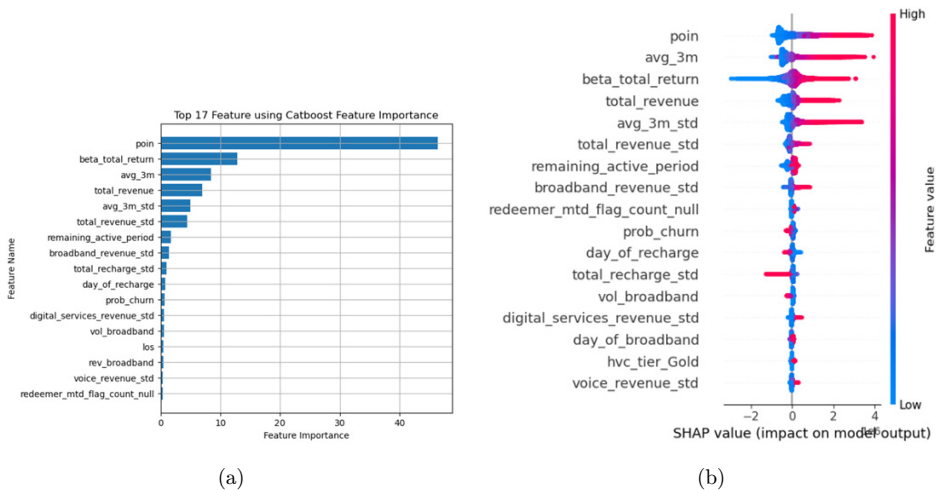


Fig. 15. Approach B — Explainability with (a) feature importance and (b) SHAP value.

explanation’s top five most essential features, consistent with Approach A. The main feature for study was “beta_total_return”, with the other top five features as auxiliary features. The dependent charts for these relationships are in Fig. D.15.

Dependence plots show that beta and SHAP are positively correlated. The charts show that beta levels increase SHAP values. Remember that SHAP and RAR values differ. The SHAP value tends to converge to zero as the beta value approaches zero, however positive beta values do not necessarily mean positive SHAP values. In contrast, positive beta values always provide larger SHAP values than income. Interestingly, the beta value and other variables interact similarly to the probability of churn in the previous approach. In particular, examples with high beta and “poin” values have greater SHAP values than those with low beta and “poin” values. However, unlike the previous strategy, there appears to be less secondary feature data with high values. This shows that the secondary feature’s high values are confined within a narrower beta range.

To evaluate strategy B’s best model, multiple splitting procedures were used. Table 15 shows 50:50 to 90:10 splitting ratios. The data show stable performance across splitting ratios, with 80:20 performing best. These results show that the model is resilient and stable across split ratios, proving its capacity to generalize to new data.

Table 16 shows Approach B five-fold CV outcomes. As in Approach A, the training and test sets show reasonable consistency across all performance indicators across the five folds. Overfitting is prevented by the model’s consistent performance on unseen and training data. Performance measures show small changes between

Table 15. Approach B — Final model performances across different split strategies.

Split ratio	MAPE	MAE	RMSE	R^2
50:50	18.84	429,117	675,870	0.91
60:40	18.41	427,052	672,814	0.91
70:30	18.32	422,827	666,257	0.91
80:20	18.32	421,986	665,482	0.91
90:10	18.45	423,016	672,573	0.91

Table 16. Approach B — Final model 5-fold CV results.

Fold	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	14.75	336,476	551,978	0.9	18.41	421,542	663,835	0.9
2	14.68	335,076	550,289	0.9	18.29	423,410	667,693	0.9
3	14.77	337,057	553,040	0.9	18.08	419,802	661,276	0.9
4	14.65	337,731	555,873	0.9	18.46	421,160	661,286	0.9
5	14.71	335,086	549,399	0.9	18.5	425,654	674,963	0.9
Avg	14.71	336,285	552,116	0.90	18.35	422,314	665,811	0.90

Table 17. Approach B — Final model group-split results.

Test group	Train				Test			
	MAPE	MAE	RMSE	R^2	MAPE	MAE	RMSE	R^2
1	14.76	328,323	541,199	0.94	18.42	493,205	751,950	0.91
2	14.31	334,576	551,253	0.94	18.29	402,007	638,864	0.91
3	14.09	337,393	554,664	0.94	21.66	408,716	649,884	0.91
4	14.89	333,949	546,087	0.94	18.46	624,200	897,520	0.89
Avg	14.51	333,560	548,301	0.94	19.21	482,032	734,555	0.91

folds, indicating model stability. A subset of the data does not significantly affect the model's performance, demonstrating its resilience and generalizability.

Despite using diverse methodologies, Approach B's group split analysis shows a consistent trend (Table 17). Test_group 4 has the highest MAE and RMSE, while test_group 3 has the lowest MAPE. However, all indicators show stable performance for the remaining groups. This constant trend strengthens our confidence in the best model's stability and dependability across splitting strategies. Approach A also shows that training scores improve with training set size while validation scores decline. The gap between training and validation scores decreased as training size increased. This suggests the model is not overfitting because it generalizes effectively to new data.

To validate the model's output, compare influential features determined by the ML model to variables stressed in conventional calculating approaches. In Approach B, we compute RAR by multiplying the discount rate and the projected return from all consumers by their beta value, thus in traditional RAR computation, the customer's beta value is crucial, and one interesting finding of this study is that the final model likewise ranks beta as a high feature. It consistently rates among the top three features in Global Explanation using SHAP and feature significance and Local Explanation using SHAP and LIME. Figure D.15's dependence plot shows a positive link between Beta Value and SHAP Value, with higher beta values correlating to lower SHAP values. This correlation is therefore in accordance to the literature. The expected return from all consumers is -1.3% , hence a negative beta value yields a positive discount rate and a positive beta value a negative discount rate. Positive discount rates result in lower RAR values, whereas negative discount rates result in higher RAR values than revenue.

4.7. Summary of findings and business and management implications

The main findings can be summarized in three groups as follows:

- Predictive Power of Models:
 - The churn model maintains comparable performance when using only the top 17 features, with a marginal decrease in accuracy across all ML algorithms tested.

- Among churn prediction models, the XGBoost classifier demonstrates superior accuracy and *F1*-score compared to other alternatives.
- For predicting RAR value, the CatBoost regressor emerges as the best performing model across all approaches, followed closely by XGBoost, while RF consistently underperforms.
- Comparing model performance with top 150 features versus top 17–18 features for RAR prediction reveals minimal differences, with MAPE deteriorating by less than 0.5%.

- **XAI Analysis:**

- The feature “poin” consistently emerges as the most significant factor in explaining RAR calculation across all approaches using both SHAP values and Feature Importance.
- In Approach A, the probability of churn is highlighted as a crucial feature in both global and local explanations, indicating its importance in predicting RAR.
- In Approach B, *beta_value* is identified as a significant feature in both global and local explanations, indicating its relevance in RAR calculation.
- Interaction patterns are observed between churn probability, *beta_value*, and other top features, influencing SHAP values in unique ways.

- **Relevance with Traditional Models:**

- A negative correlation exists between the probability of customer churn and the predicted RAR value, aligning with traditional calculation methods.
- Beta value in Approach B shows a positive correlation with the predicted RAR value, consistent with traditional methods.
- Both the probability of churn in Approach A and the *beta_value* in Approach B, identified as highly significant features, align with traditional RAR calculations.
- Features such as *avg_3m*, *avg_3m_std*, *total_revenue*, and *total_revenue_std* consistently appear among the top features for both Approaches A and B, reaffirming their relevance in RAR calculation.

These findings collectively enhance our understanding of the factors influencing churn predictions and RAR calculations. They provide valuable insights for developing more accurate and robust models, enabling businesses to optimize decision-making processes and mitigate revenue loss risks effectively. They also hold significant implications for business management strategies, particularly in the domains of churn prediction and RAR calculation. Understanding these implications can aid businesses in devising more effective approaches to mitigate revenue loss risks and optimize profitability.

- **Streamlined Feature Selection for Model Development:** The study highlights the efficiency of utilizing a streamlined set of features for churn prediction and RAR calculation models. By focusing on the top 17–18 most influential features, businesses can simplify model development processes without sacrificing predictive

accuracy. This streamlined approach reduces computational complexity and resource requirements, enabling faster model deployment and facilitating more agile decision-making processes.

- **Selection of Optimal Machine Learning Algorithms:** The superiority of the XGBoost classifier for churn prediction underscores the importance of selecting the most suitable ML algorithms for specific prediction tasks. Businesses can benefit from investing in algorithmic diversity and experimentation to identify the optimal model architecture for their unique requirements. Additionally, understanding the strengths and weaknesses of different algorithms can inform strategic decisions regarding model selection and deployment.
- **Targeted Risk Mitigation Strategies:** Insights from XAI analysis shed light on the key factors driving churn prediction and revenue risk. By identifying significant features such as churn probability and beta_value, businesses can develop targeted risk mitigation strategies to address customer churn and optimize revenue streams. These strategies may include personalized retention campaigns, pricing adjustments, or product/service enhancements tailored to high-risk customer segments.
- **Leveraging Interaction Patterns for Enhanced Insights:** The observed interaction patterns between churn probability, beta_value, and other top features provide deeper insights into the complex dynamics influencing revenue risk. Businesses can leverage these insights to gain a more nuanced understanding of customer behavior and its impact on revenue generation. By analyzing these interaction patterns, organizations can uncover hidden correlations and dependencies, enabling more informed decision-making and proactive risk management.
- **Alignment with Traditional Calculation Methods:** The consistency between the developed models and traditional RAR calculation methods validates their relevance and applicability in real-world business contexts. Businesses can leverage these models as complementary tools to existing risk management frameworks, enhancing their ability to anticipate and mitigate revenue loss risks. Moreover, the identification of consistent features such as avg_3m and total_revenue reaffirms their importance in traditional RAR calculations, providing businesses with actionable insights to optimize revenue performance.

Therefore, the findings from the analysis offer actionable insights for businesses seeking to enhance churn prediction and revenue risk management strategies. By leveraging streamlined feature selection techniques, selecting optimal ML algorithms, and harnessing insights from XAI analysis, organizations can develop more accurate and tailored approaches to mitigate revenue loss risks and optimize business outcomes. Additionally, the alignment with traditional calculation methods and the identification of consistent features further validate the relevance and applicability of the developed models in real-world business scenarios. Overall, these implications underscore the importance of leveraging advanced analytics and ML techniques to drive strategic decision-making and enhance business performance.

5. Conclusion

This research marks a significant milestone in advancing the understanding of customer value within the telecommunications industry by introducing and implementing a novel approach to assessing customers' value through RAR. By integrating both potential revenue and associated risks into the evaluation process, this study offers a comprehensive perspective on customer worth, filling a notable research gap and expanding the breadth of existing studies on RAR.

5.1. *Lessons learned and contributions*

The practical implications of this research are profound, offering telecommunications companies a more nuanced understanding of their customers' value and enabling data-driven, tailored customer treatments and portfolio management strategies. By providing insights into customer churn probability and revenue volatility, the developed models empower companies to optimize profits and drive business strategies more efficiently.

The contributions of this research extend beyond practical applications to theoretical advancements in several key areas. First, this study pioneers a focus on calculating and predicting RAR specifically within the telecommunications industry, offering an original perspective that enriches the broader discourse on RAR. By situating RAR within the unique context of the telecommunications sector, this research expands the methodological toolkit available to researchers and practitioners, fostering a deeper understanding of customer-related risks in this industry.

Second, this research contributes to the existing literature on integrating risk into customer valuation by introducing novel methods for calculating RAR. These methods, grounded in ML techniques, provide innovative approaches to capturing and quantifying customer-related risks, thereby enhancing the accuracy and robustness of customer value assessments.

Moreover, the application of ML for RAR prediction in this study represents a significant scientific development. While previous studies have explored risk-adjusted CLV, only a few have fully leveraged ML to predict RAR. By demonstrating the practical application of ML techniques in predicting RAR, this research not only underscores the potential of ML in this context but also sets a precedent for future research on RAR prediction.

Finally, this study contributes to the field of XAI by leveraging XAI techniques to assess feature importance in RAR modeling. By comparing the results of XAI analysis with traditional calculation methods, this research highlights the potential of XAI for delivering clearer, more understandable ML outcomes, thereby advancing our understanding of the factors influencing RAR.

5.2. Limitations and future research

Despite its contributions, this research is not without limitations. One significant limitation is the selection of risk parameters, which primarily focuses on the probability of customer churn and revenue volatility. Future research could explore a broader range of risk factors inherent in the telecommunications sector, such as geographical variances and shifting customer segments, to provide a more comprehensive understanding of customer behavior.

Additionally, the methodology adopted in this study, while comprehensive, may benefit from further refinement. For example, future research could explore more advanced ML models and optimization techniques to improve predictive accuracy and model performance. Moreover, incorporating advanced methods for assessing feature importance, such as causal and asymmetric SHAP values, could provide a more nuanced understanding of the effects of different features on RAR.

Furthermore, future research could delve into the implications of RAR for different segments of the telecommunications industry, such as prepaid versus postpaid customers or consumer versus enterprise markets. Understanding how RAR varies across these segments could provide valuable insights for tailored marketing strategies and portfolio management approaches.

In conclusion, while this research represents a significant advancement in the understanding of RAR in the telecommunications industry, there are ample opportunities for future research to explore and refine the methodologies, models, and applications introduced in this study. By addressing these limitations and building upon the findings of this research, future studies can continue to enrich our understanding of RAR and its implications for businesses in the telecommunications sector.

Appendix A. Data Dictionary — Features Description

Table A.1. Sample data dictionary.

Feature name	Data type	Description
msisdn	Int	The mobile phone number associated with the customer.
poin	Int	The loyalty points earned by the customer.
total_revenue	Int	The total revenue generated by the customer.
total_recharge	Int	The total recharge conducted by the customer.
dominan_denomination	Int	The dominant or most frequent recharge denomination used by the customer.
Dominan_recharge_channel	Obj	The dominant channel used by the customer to recharge.
avg_3m	Int	The average of last three months revenue by the customer.
voice/sms/broadband/ digital_services	int	The type of services used by the customer.
revenue_xxx	Int	The revenue generated by using the service “xxx”.

Table A.1. (Continued)

Feature name	Data type	Description
transaction_xxx	Int	The total number of transactions for the service “xxx” made by the customer.
duration_xxx	Int	The total duration of “xxx” made by the customer.
volume_of_xxx	Int	The total volume of “xxx”.
total_day_of_xxx	Int	The total number of days customer using “xxx” service.
xxx_package	Int	The package related for service purchased by customer.
last_recharge/revenue	Datetime	The last recorded date of such transaction.
xxx_std	Int	Indicate the standard deviation of such value over the observation period.
xxx_pct_change	Int	Indicate the average change (in pct) of such value over the observation period.
status	Obj	The status of the customer’s account or subscription.
kabupaten	Obj	The name of the district or regency where the customer is located.
kelurahan	Obj	The name of the sub-district or neighborhood where the customer is located.
dominan_revenue	Obj	The dominant or most frequent revenue type or category for the customer.
gender	Obj	The gender of the customer.
Brand_x	Obj	The pricing plan/brand selected by customer.
region_lacci_xxx	Obj	The region or location identifier based on the LAC-CI (Location Area Code-Cell Identity).
los	Int	The length of stay or tenure of the customer.
device_type	Int	The type or model of the customer’s device.
Device_data_capable	Int	Indicates whether the customer’s device is capable of using data services.
Device_os_priority	Int	The priority or preference given to the customer’s operating system.
Device_network_prio	Int	The priority or preference given to the customer’s network connection.
data_usage_category	Int	The category or classification of the customer based on their data usage.
Most_used_apps	Int	The name of the app ranked first by the customer.

Appendix B. Hyperparameters

Table B.1. Hyperparameters used for each model.

Model	Algorithm	Hyperparameter
Churn	LR	max_iter = 1000, penalty = “l1”, C = 0.01, solver = “liblinear”
	RF	n_estimators = 500, min_samples_split = 10, min_samples_leaf = 2, max_features = “auto”, max_depth = None,
	XGBoost	subsample = 0.8, n_estimators = 500, max_depth = 6, learning_rate = 0.05, gamma = 0.1, colsample_bytree = 0.8,
	CatBoost	n_estimators = 1000, learning_rate = 0.05, depth = 8.
	RF	bootstrap = True, max_depth = None, max_features = auto, min_samples_leaf = 3, min_samples_split = 2, n_estimators = 500

Table B.1. (Continued)

Model	Algorithm	Hyperparameter
RAR — beta value approach	XGBoost	colsample_bytree = 1.0, gamma = 0, learning_rate = 0.05, max_depth = 9, n_estimators = 1000, subsample = 1
	CatBoost	Depth = 9, learning_rate = 0.15, loss_function = MAE, n_estimators = 800.
	RF	Bootstrap = True, max_depth = None, max_features = auto, min_samples_leaf = 2, min_samples_split = 2, n_estimators = 300
RAR — WACC and churn approach	XGBoost	colsample_bytree = 1.0, gamma = 0, learning_rate = 0.05, max_depth = 9, n_estimators = 900, subsample = 0.8
	CatBoost	n_estimators = 1100, depth = 9, learning_rate = 0.08, loss_function = MAE.

Appendix C. Probability of Customer Churn

Table C.1. Probability of customer churn: Model performance (5-fold CV).

Model and model development stage	Accuracy	Recall	Precision	F1-score	AUC
Logistic regression					
Initial model and all features	0.8	0.77	0.81	0.79	0.8
Feature selection	0.79	0.7	0.86	0.77	0.79
Grid search hyperparameter tuning	0.79	0.7	0.86	0.77	0.79
CatBoost					
Initial model and all features	0.853	0.832	0.87	0.85	0.853
Feature selection	0.852	0.828	0.869	0.848	0.852
Random search hyperparameter tuning	0.854	0.847	0.859	0.853	0.854
Grid search hyperparameter tuning	0.856	0.844	0.865	0.854	0.856
XGBoost					
Initial model and all features	0.89	0.886	0.894	0.89	0.89
Feature selection	0.889	0.884	0.893	0.889	0.889
Random search hyperparameter tuning	0.888	0.887	0.89	0.888	0.888
Grid search hyperparameter tuning	0.892	0.89	0.893	0.892	0.892
Random Forest					
Initial model and all features	0.799	0.722	0.854	0.782	0.799
Feature selection	0.796	0.695	0.871	0.773	0.796
Random search hyperparameter tuning	0.789	0.653	0.896	0.756	0.789
Grid search hyperparameter tuning	0.797	0.694	0.875	0.774	0.797

Appendix D. Customer churn and RAR

Table D.1. Approach A — Descriptive statistic of calculated RAR result.

Metric	Total revenue	RAR value		
		WACC min	WACC mid	WACC max
count	197,692	197,692	197,692	197,692
mean	2,683,918	2,255,263	2,195,887	2,190,303
std	2,144,940	1,855,696	1,823,163	1,820,201
min	2,550	1,697	1,588	1,577

Table D.1. (Continued)

Metric	Total revenue	RAR value		
		WACC min	WACC mid	WACC max
25%	1,110,838	903,632	870,937	867,880
50%	2,113,714	1,738,899	1,682,397	1,676,812
75%	3,580,973	3,024,572	2,948,474	2,940,945
max	17,724,060	10,821,540	10,821,000	10,820,950

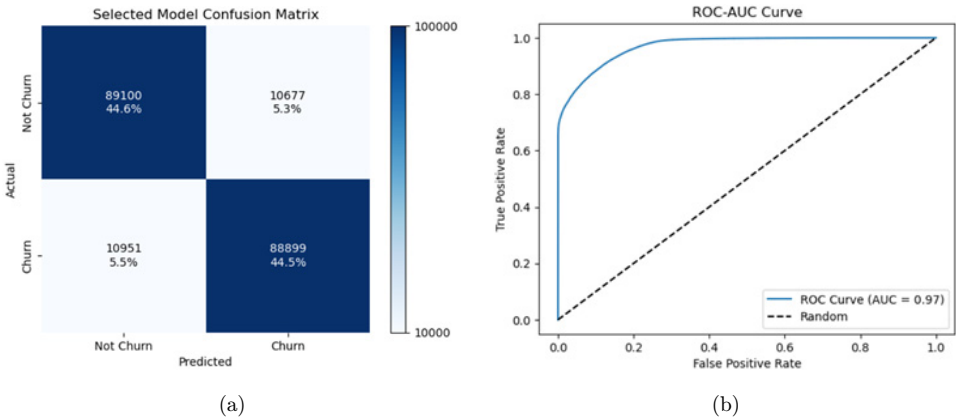


Fig. D.1. Customer churn prediction model: (a) Confusion matrix and (b) ROC–AUC curve.

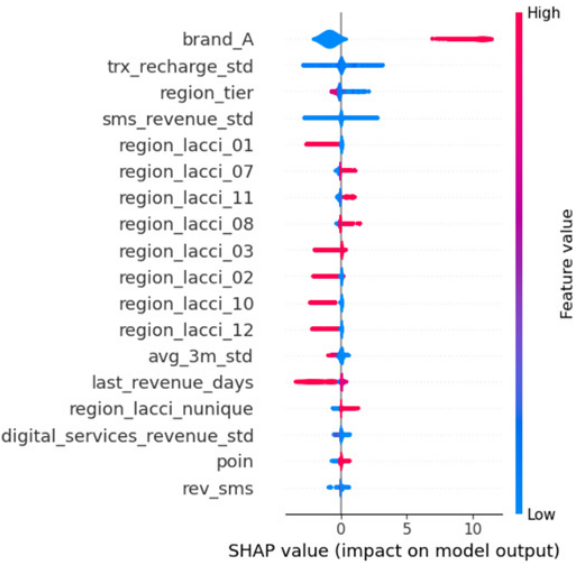


Fig. D.2. Customer churn prediction model: SHAP value.

Table D.2. Approach B — Descriptive statistics of calculated RAR result.

Metric	Total revenue	RAR value
count	194,464	194,464
mean	2,702,213	2,809,040
std	2,164,500	2,279,828
min	2,550	1,652
25%	1,117,476	1,131,818
50%	2,125,282	2,185,314
75%	3,601,637	3,765,111
max	23,360,000	13,277,820

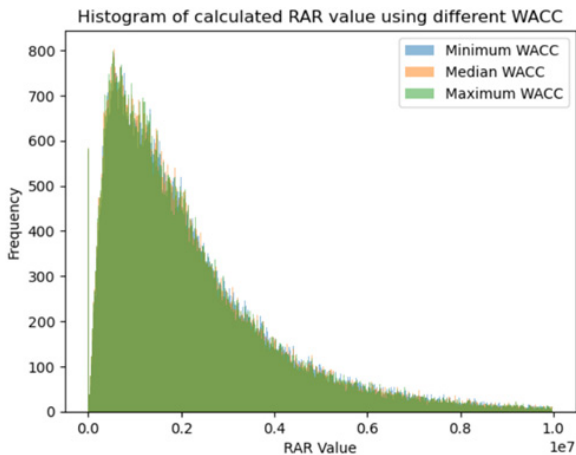


Fig. D.3. Approach A — Distributed RAR value.

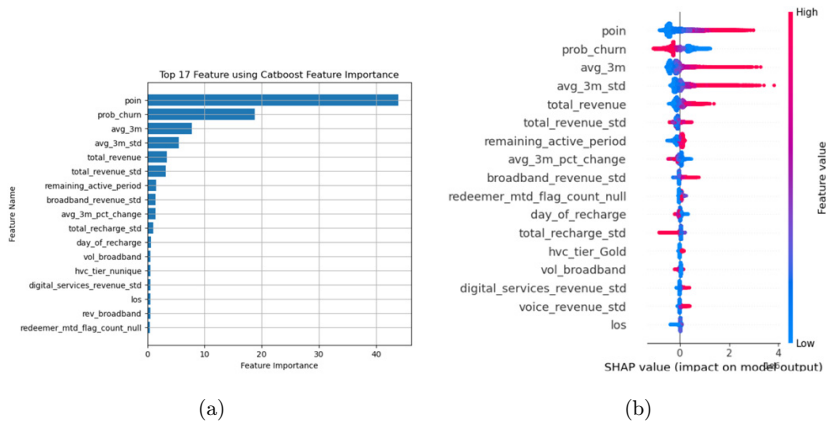


Fig. D.4. Approach A.1 — Explainability with (a) feature importance and (b) SHAP value.

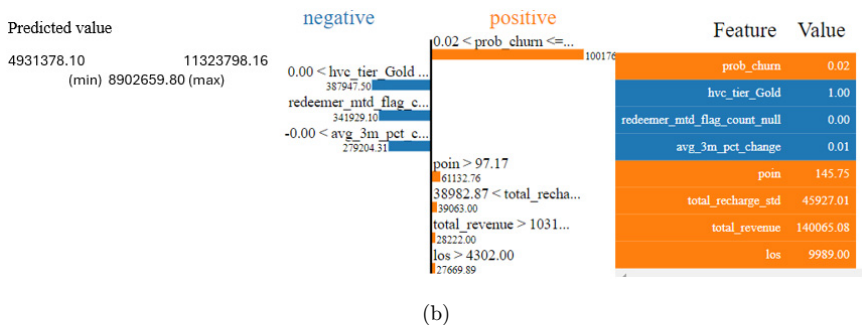
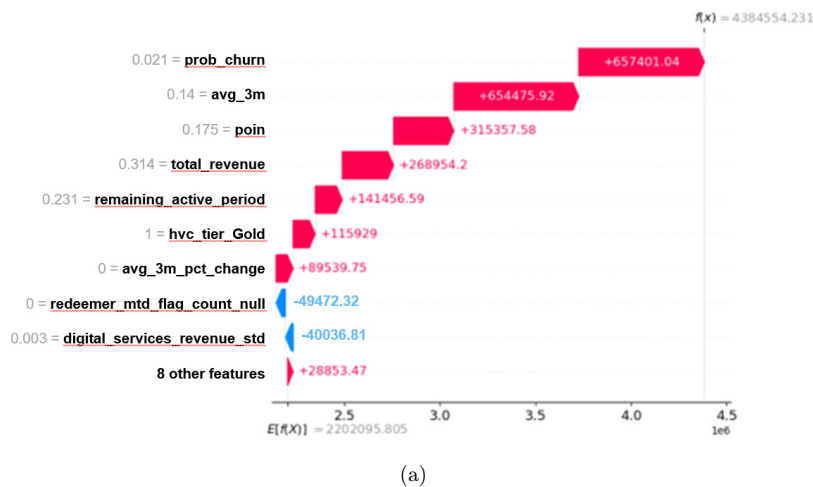


Fig. D.5. Approach A.1 — Explainability with (a) SHAP value and (b) LIME value for data sample X.

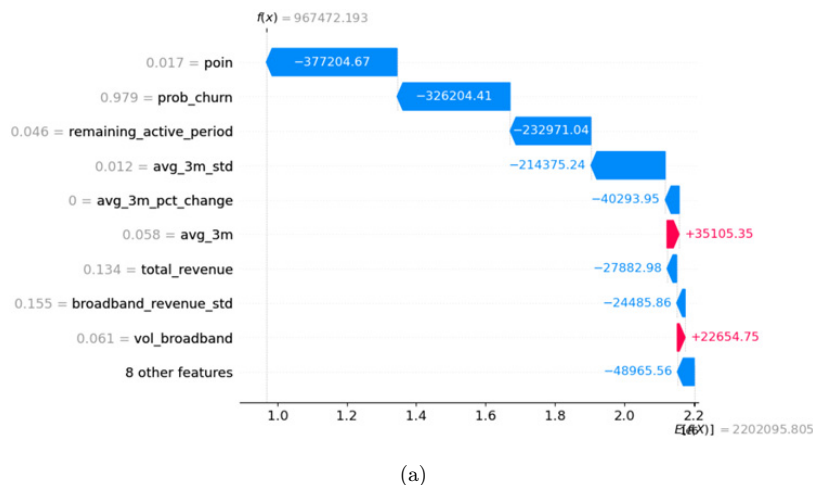


Fig. D.6. Approach A.1 — Explainability with (a) SHAP value and (b) LIME value for data sample Y.

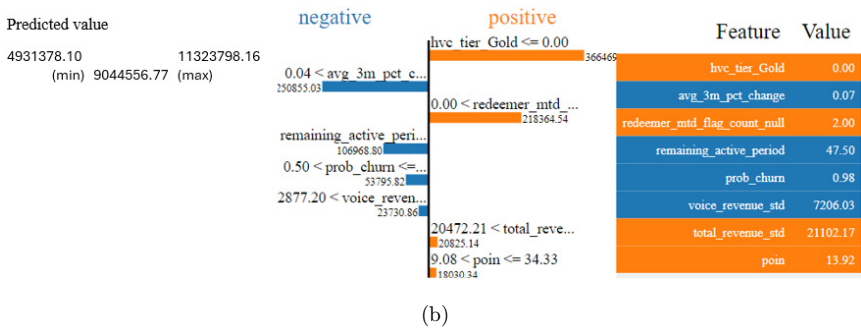


Fig. D.6. (Continued)

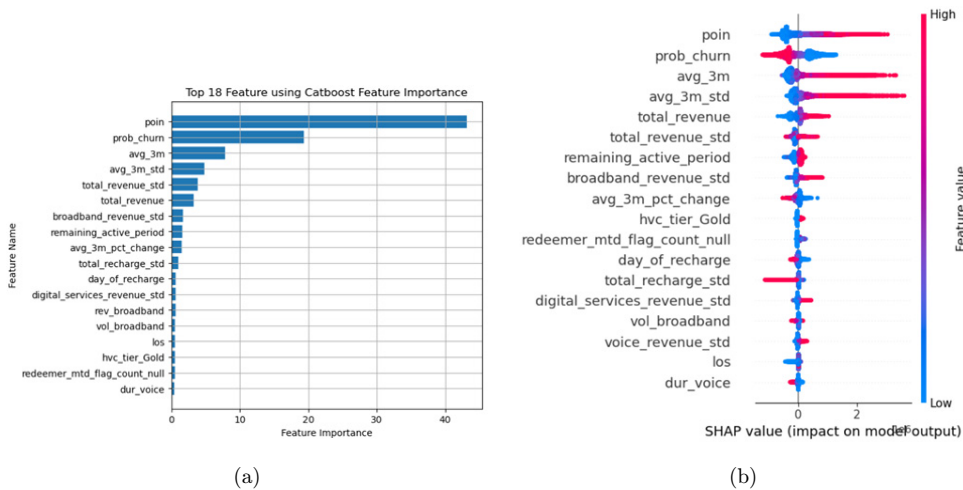
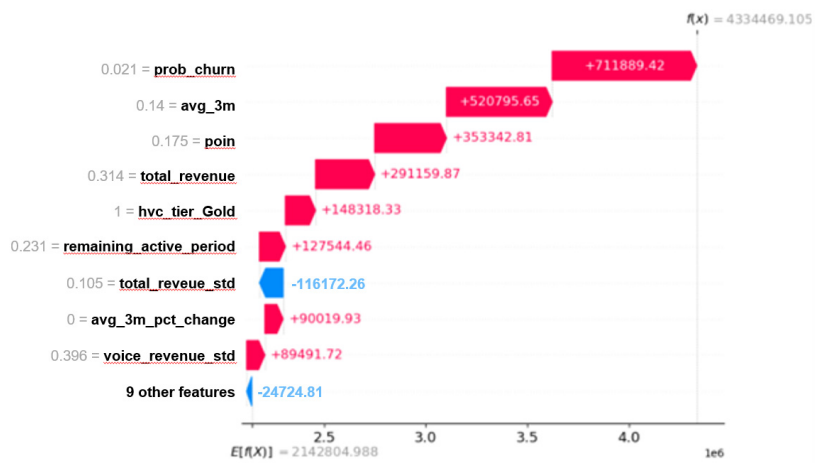
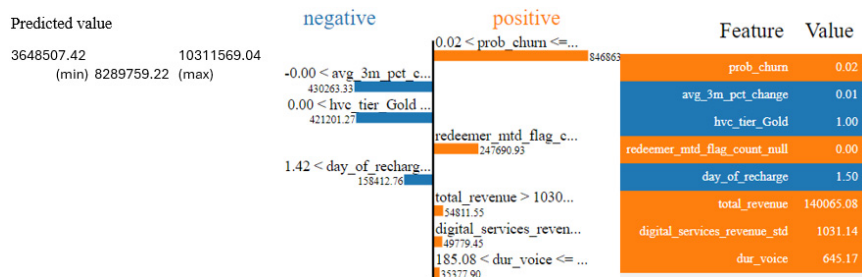


Fig. D.7. Approach A.2 — Explainability with (a) feature importance and (b) SHAP value.

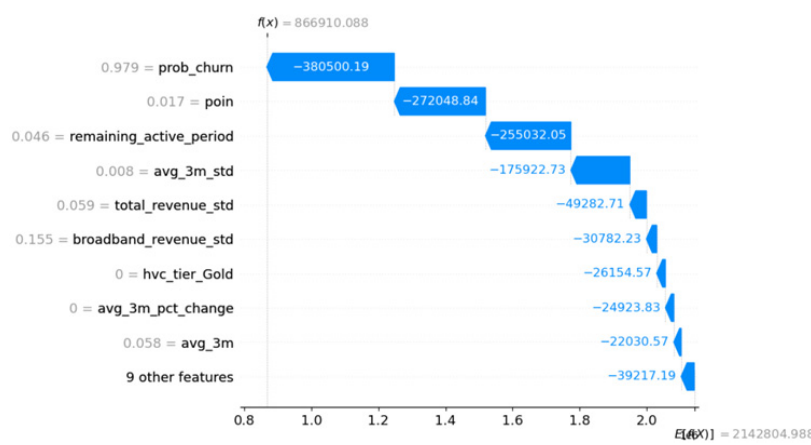


(a)



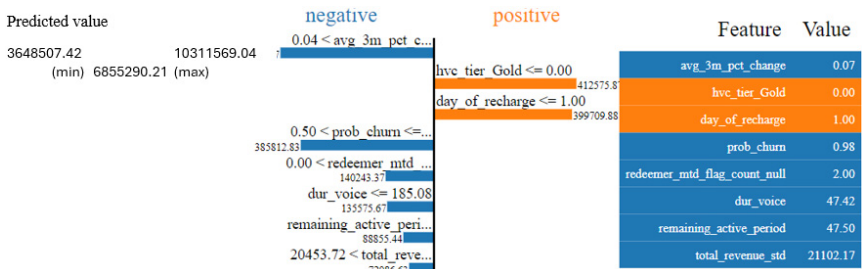
(b)

Fig. D.8. Approach A.2 — Explainability with (a) SHAP value and (b) LIME value for data sample X.



(a)

Fig. D.9. Approach A.2 — Explainability with (a) SHAP value and (b) LIME value for data sample Y.



(b)

Fig. D.9. (Continued)

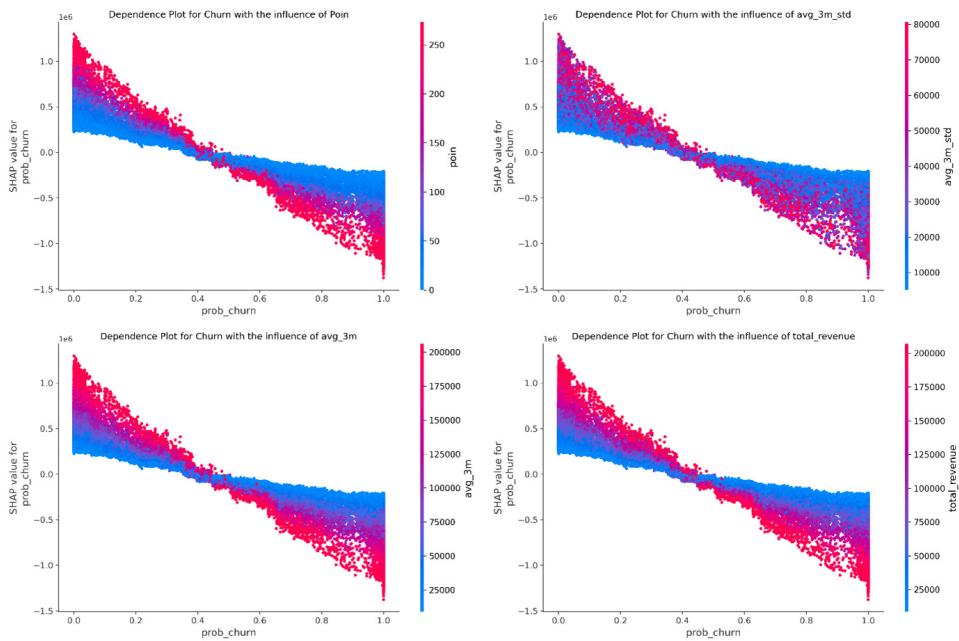
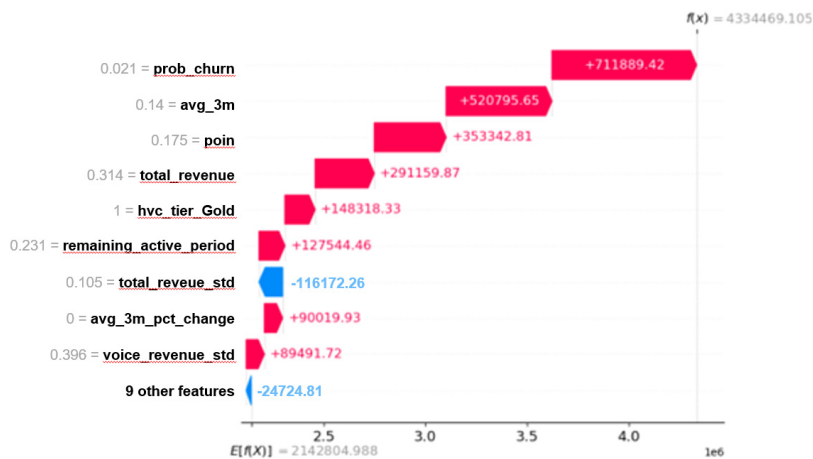
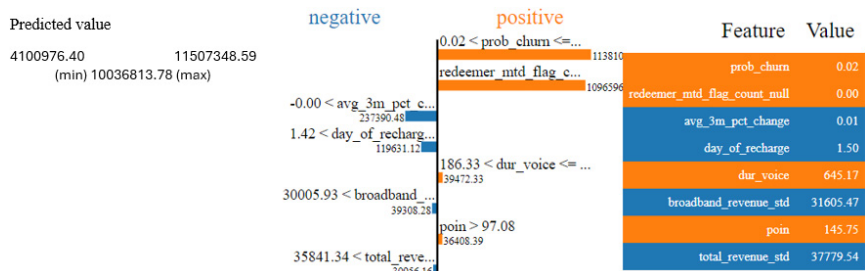


Fig. D.10. Approach A.2 — SHAP dependence plot of probability of churn.

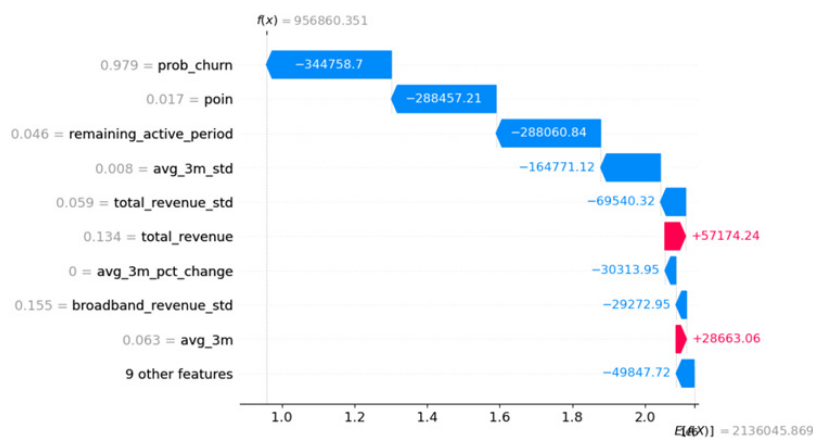


(a)



(b)

Fig. D.11. Approach A.3 — Explainability with (a) SHAP value and (b) LIME value for data sample X.



(a)

Fig. D.12. Approach A.3 — Explainability with (a) SHAP value and (b) LIME value for data sample Y.

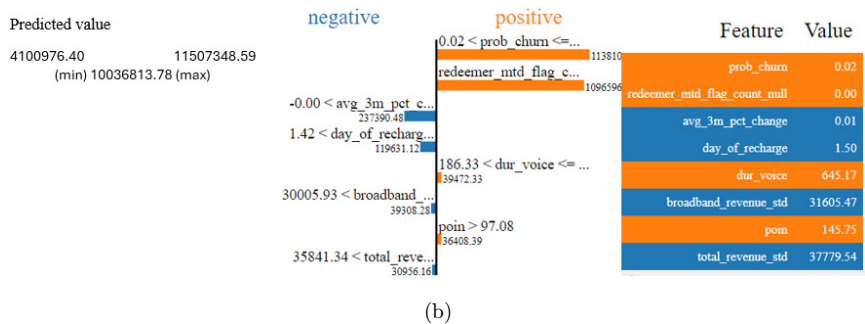


Fig. D.12. (Continued)

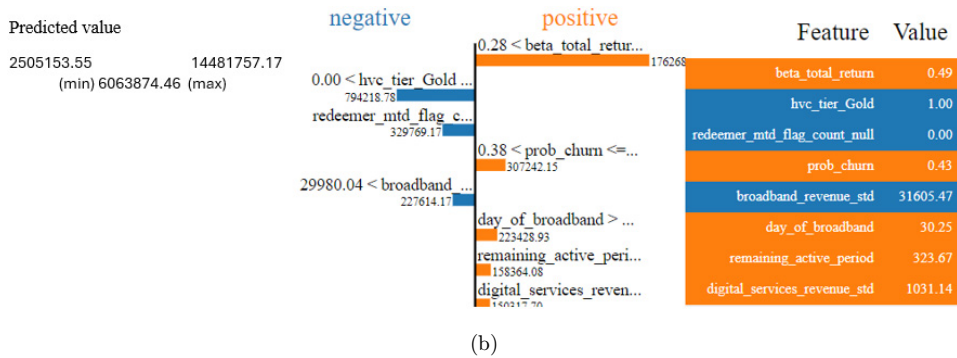
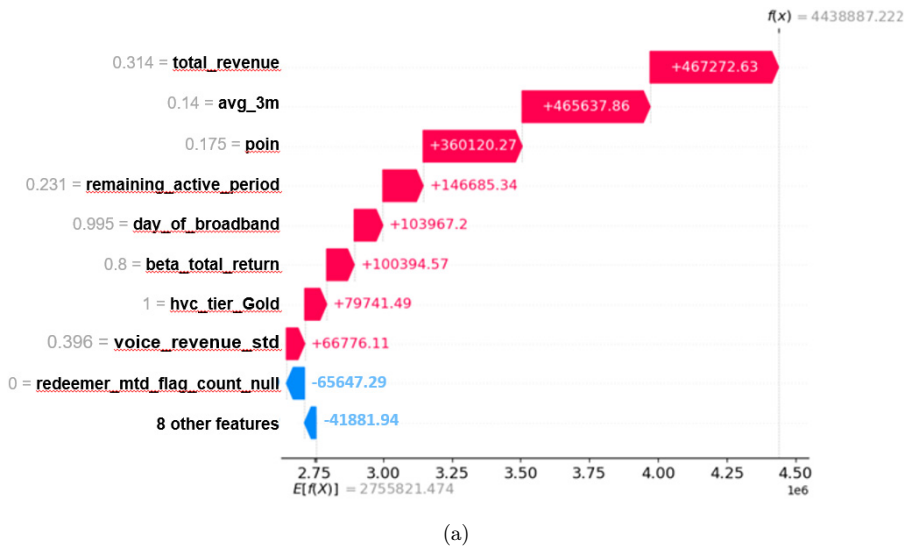


Fig. D.13. Approach B — Explainability with (a) SHAP value and (b) LIME value for data sample X.

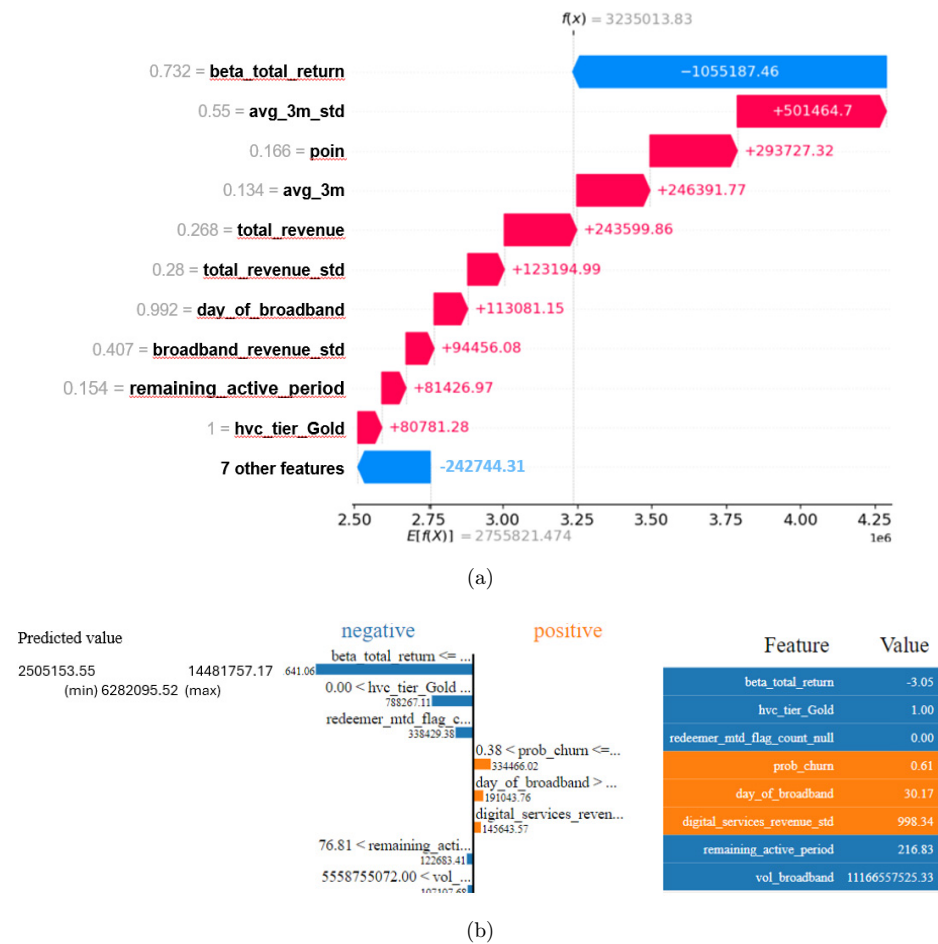


Fig. D.14. Approach B — Explainability with (a) SHAP value and (b) LIME value for data sample Y.

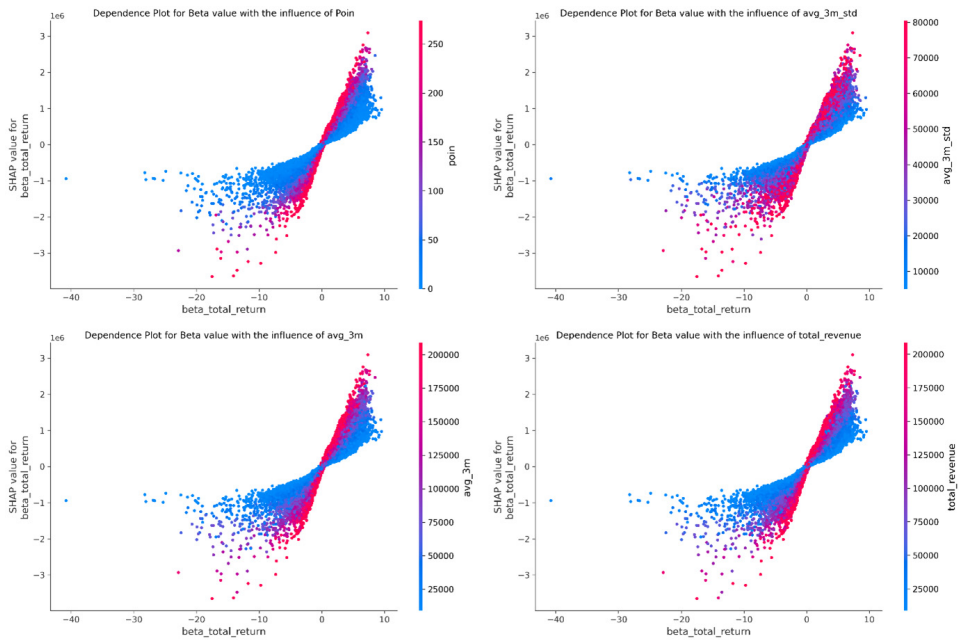


Fig. D.15. Approach B — SHAP dependence plot of probability of churn.

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